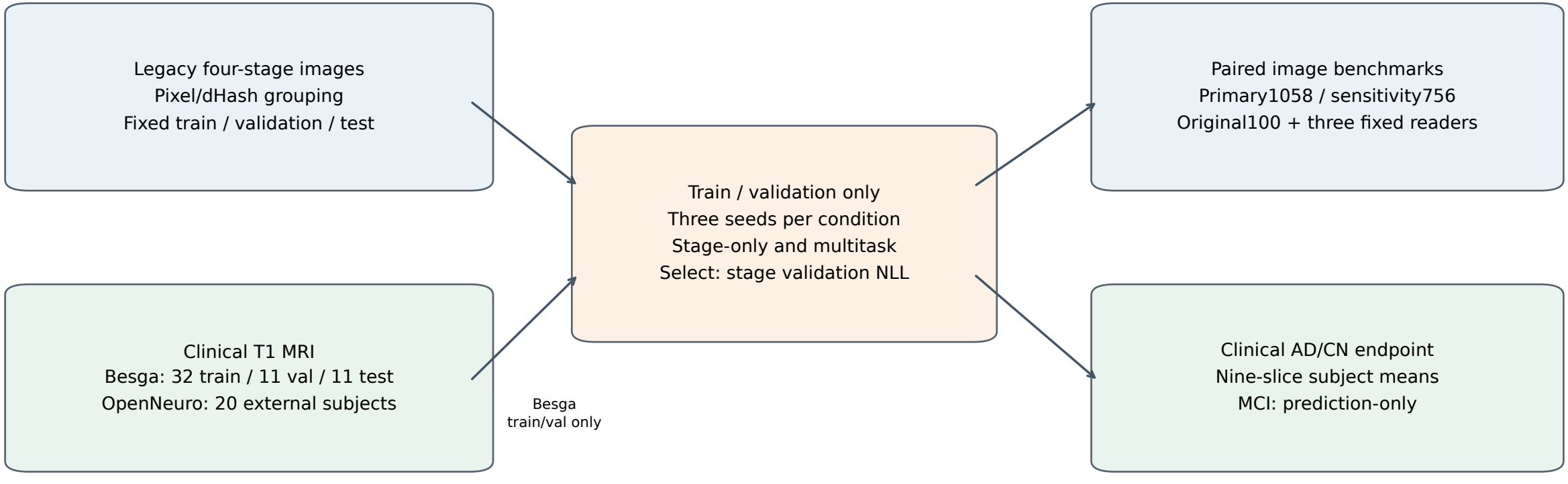
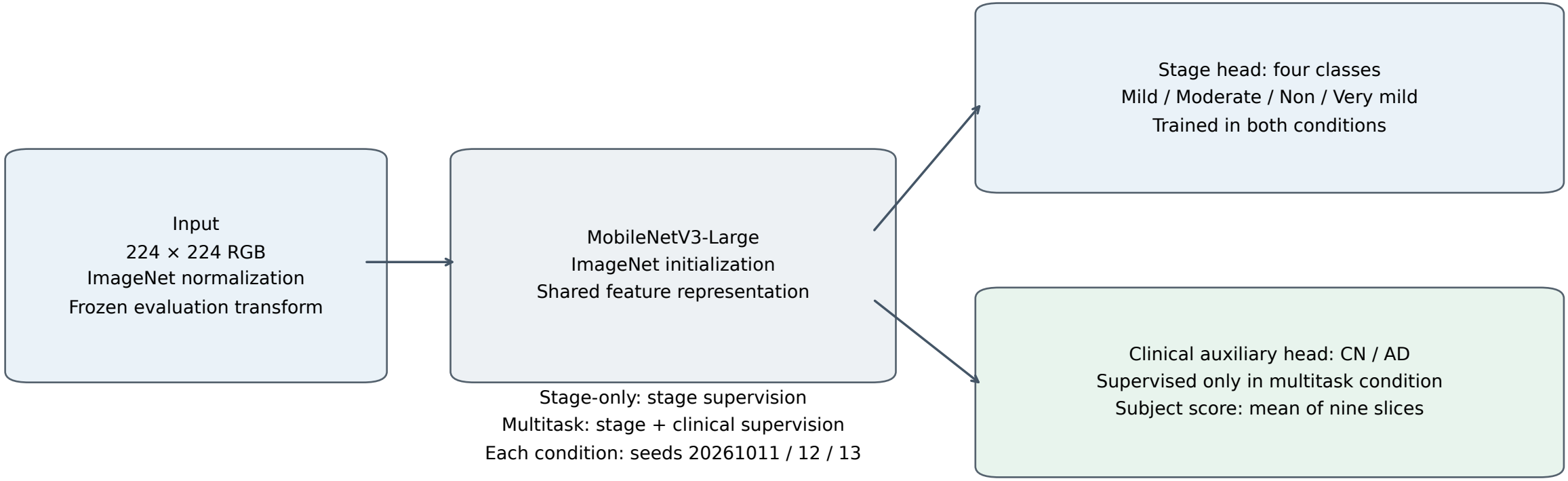
Kevin dementia study | data and evaluation workflow



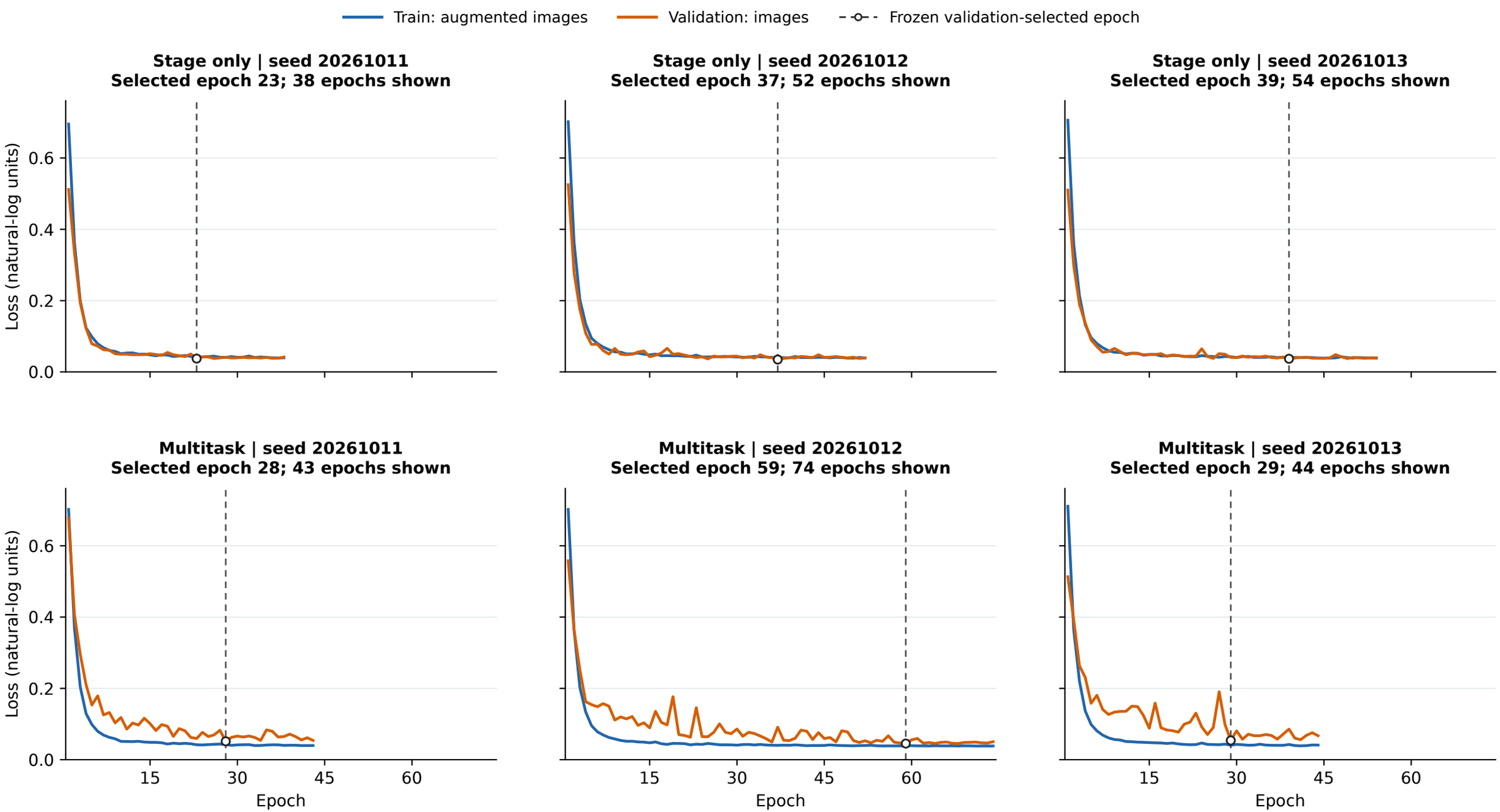
Report every frozen run and both fixed supplementary ensembles
Conditional paired intervals for images; exact intervals for clinical subjects

Test/external cases enter evaluation only. Original100 is reused; hash groups do not prove patient identity. Clinical cohorts are small and source-limited.
All evaluations remain research-only.

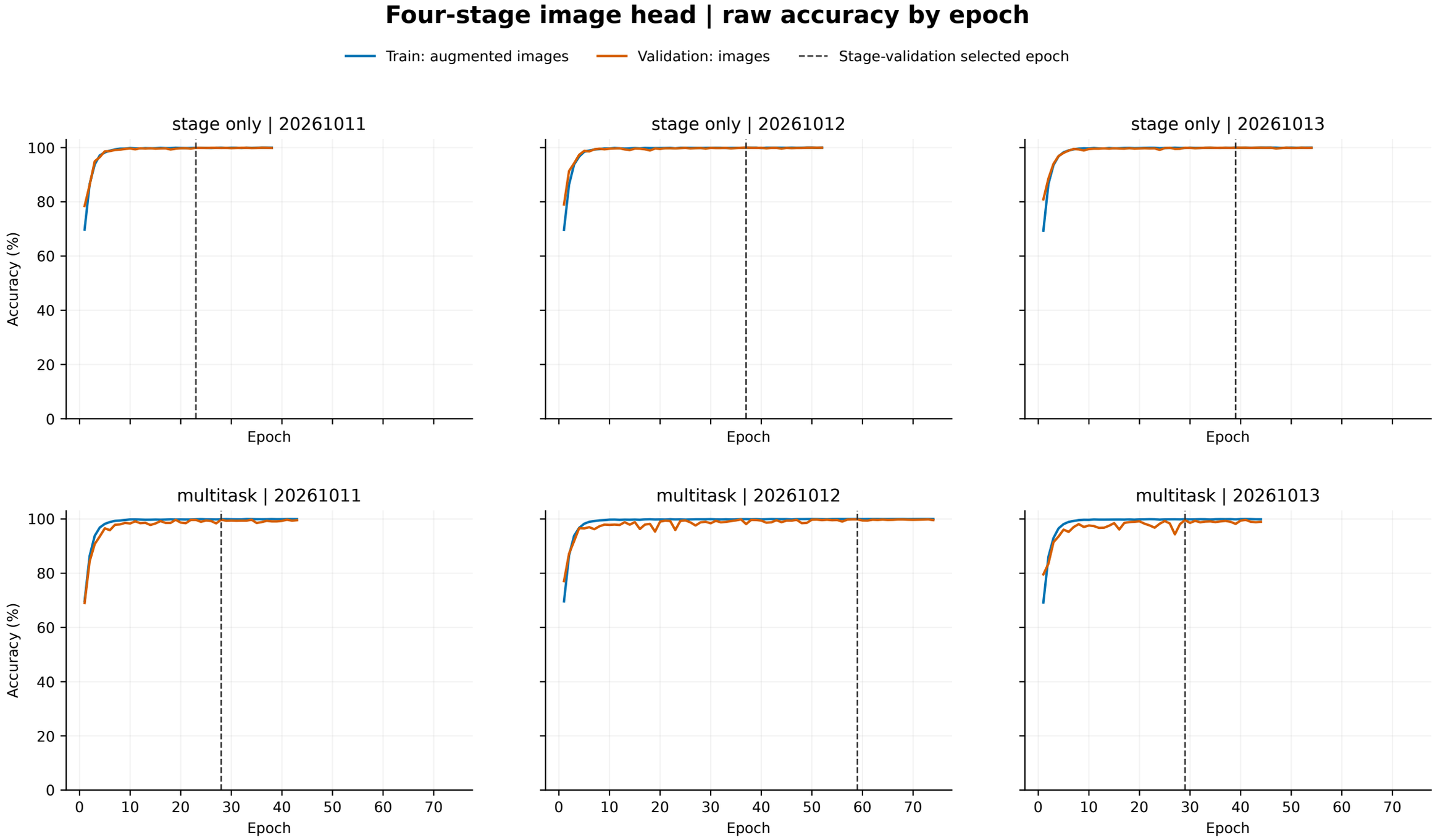
Dementia model | two prespecified training conditions



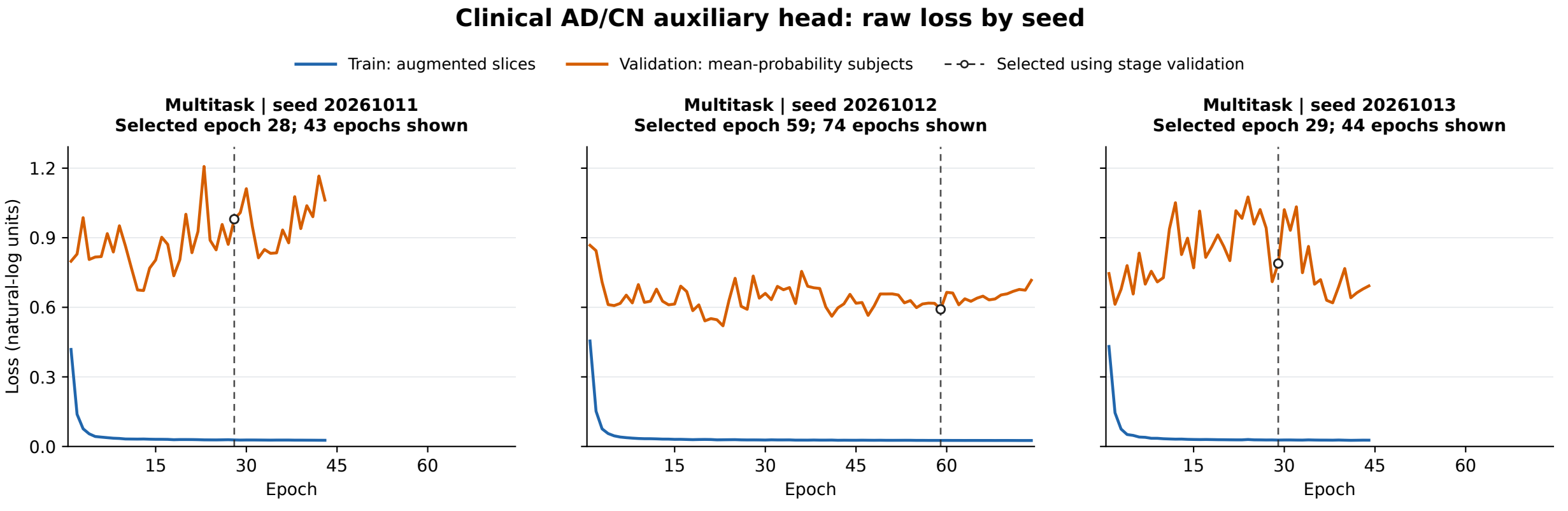
The legacy stage head and clinical AD/CN head have different reference labels and evaluation units. Stage-only auxiliary weights are not evaluated clinically. Uniform three-seed probability ensembles are supplementary postprocessing, with no fitted weights.



Every recorded epoch is shown without smoothing. Losses are the trainer's unweighted, unsmoothed-label cross-entropy / NLL summaries; training uses augmentation. These are stage-head losses, not the weighted, label-smoothed training objective or the total multitask objective.

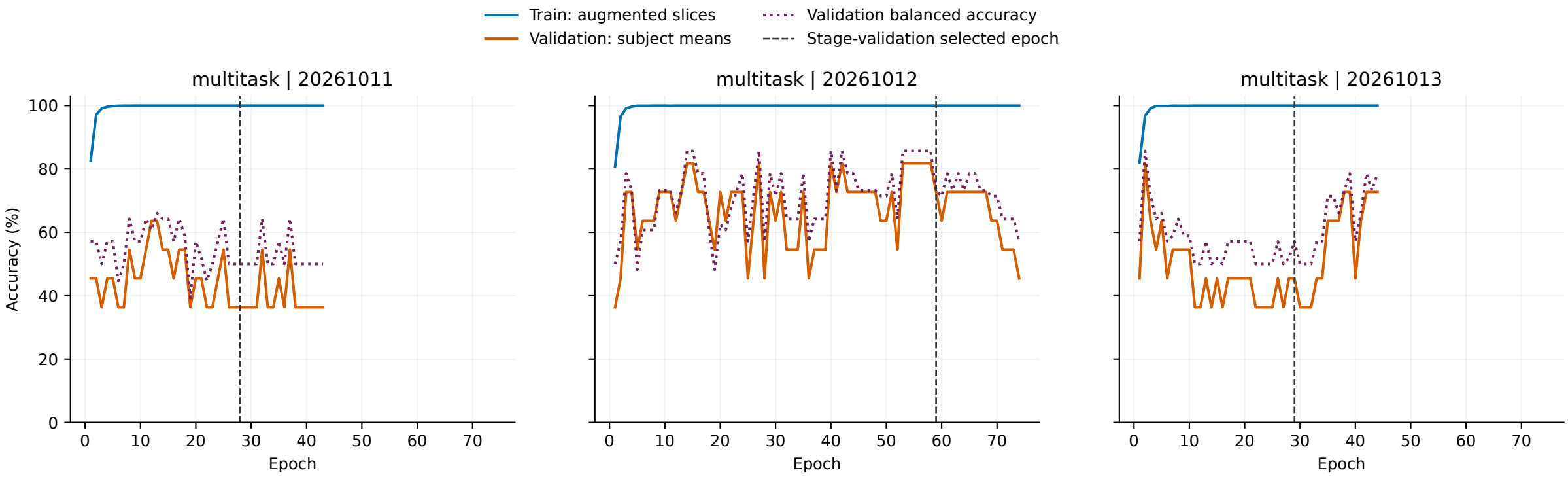


Every epoch is shown without smoothing; all six fixed runs remain visible. Dashed line is the frozen stage-validation-selected checkpoint. Curves do not establish patient-independent clinical generalization.



Every recorded epoch is shown without smoothing. Training loss uses sampled augmented slices; validation loss uses each subject's mean of nine slice probabilities.
The two curves use different aggregation units, so their gap is not directly comparable. Checkpoints were selected by four-stage validation NLL, not clinical loss.

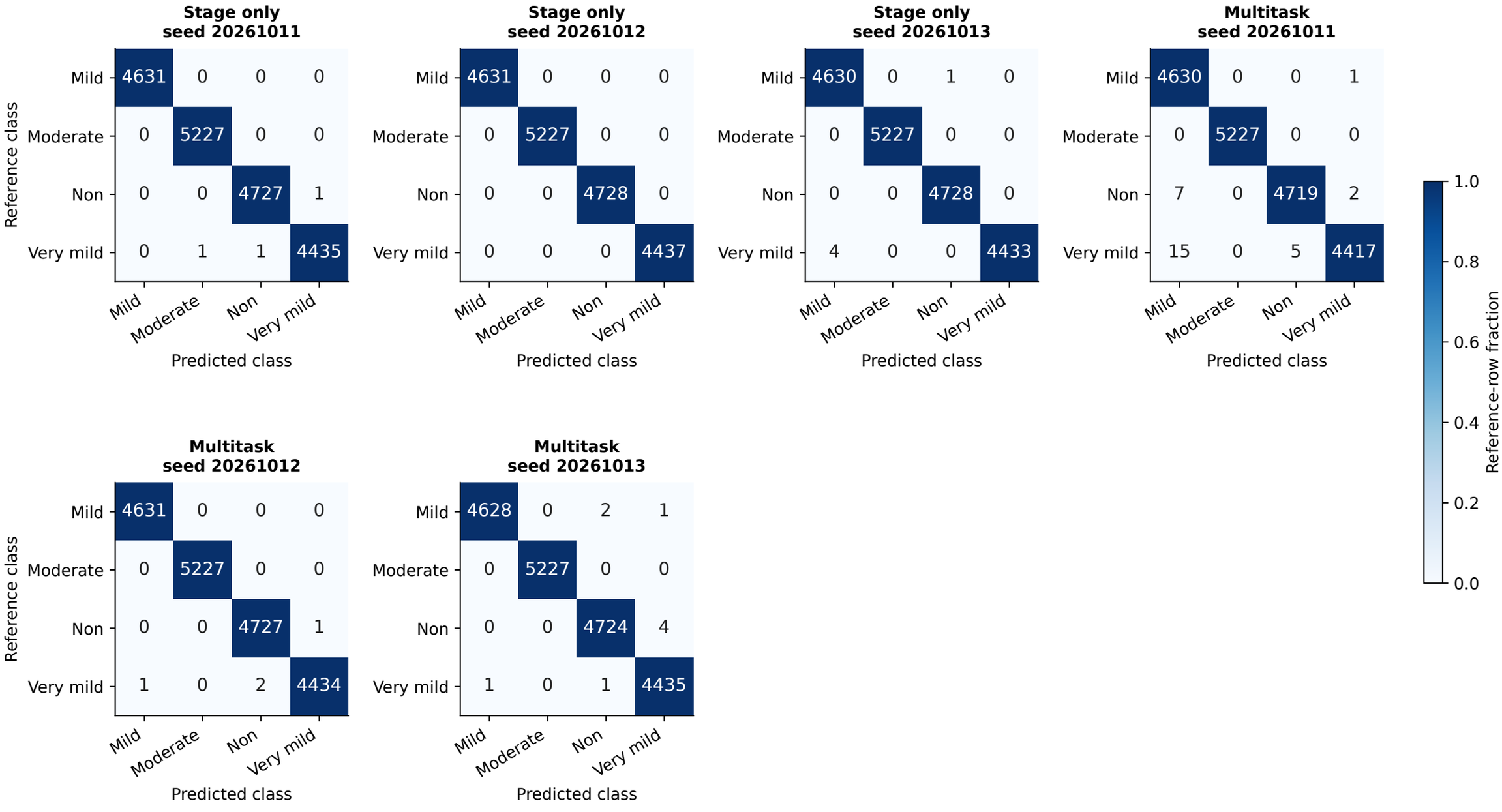
Clinical AD/CN | raw accuracy by epoch



Training uses augmented slices and validation averages nine slices per subject; these are different units. Dashed line is the frozen stage-validation-selected checkpoint. Curves do not establish patient-independent clinical generalization.

Training19023 | final-checkpoint resubstitution | n=19023

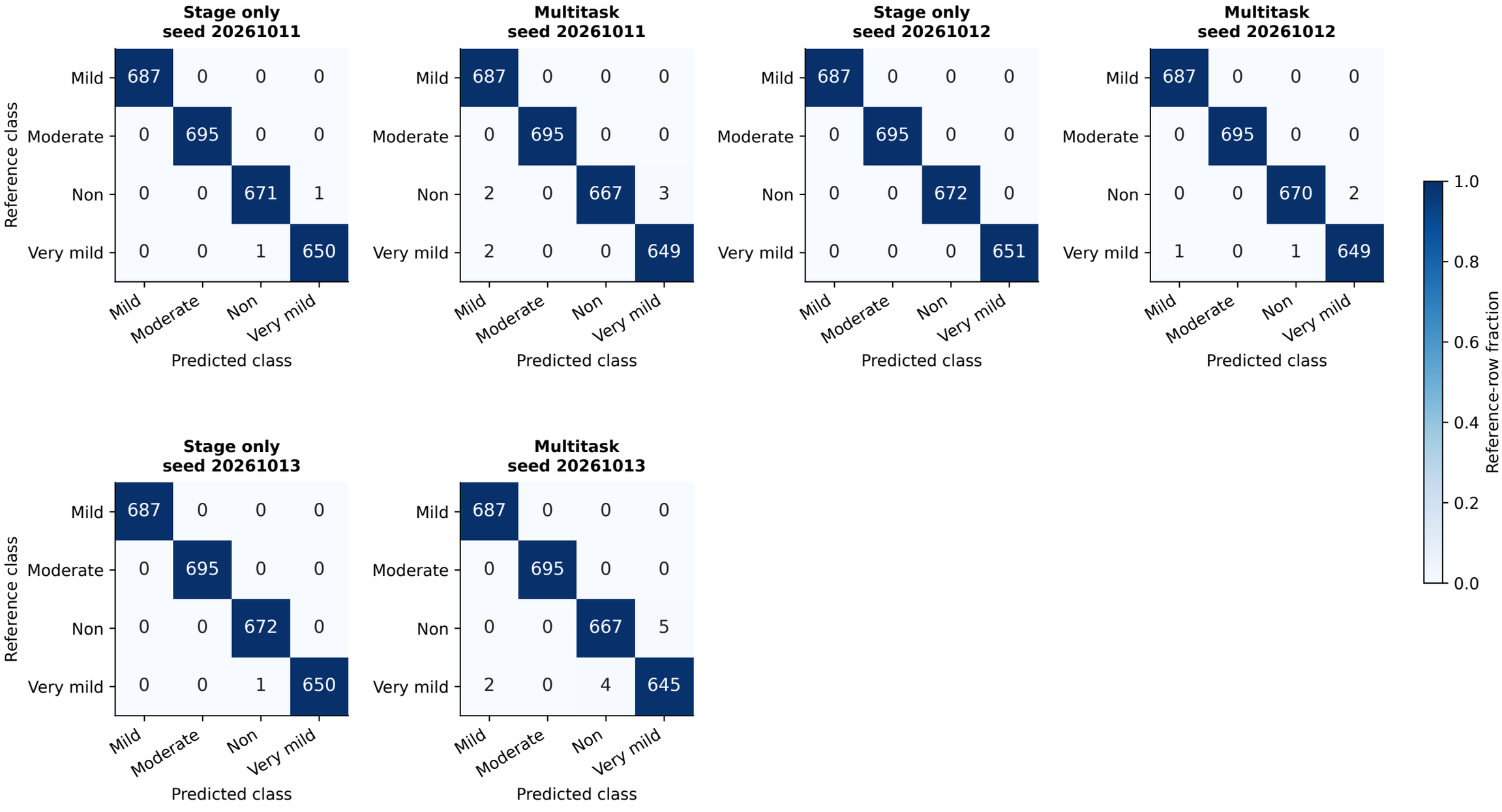
Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.



Clean final-checkpoint predictions on training images; not augmented epoch draws or generalization evidence. All six frozen candidates; no model selection. Image groups are not patient identities.

Validation at each frozen selected checkpoint | n=2705

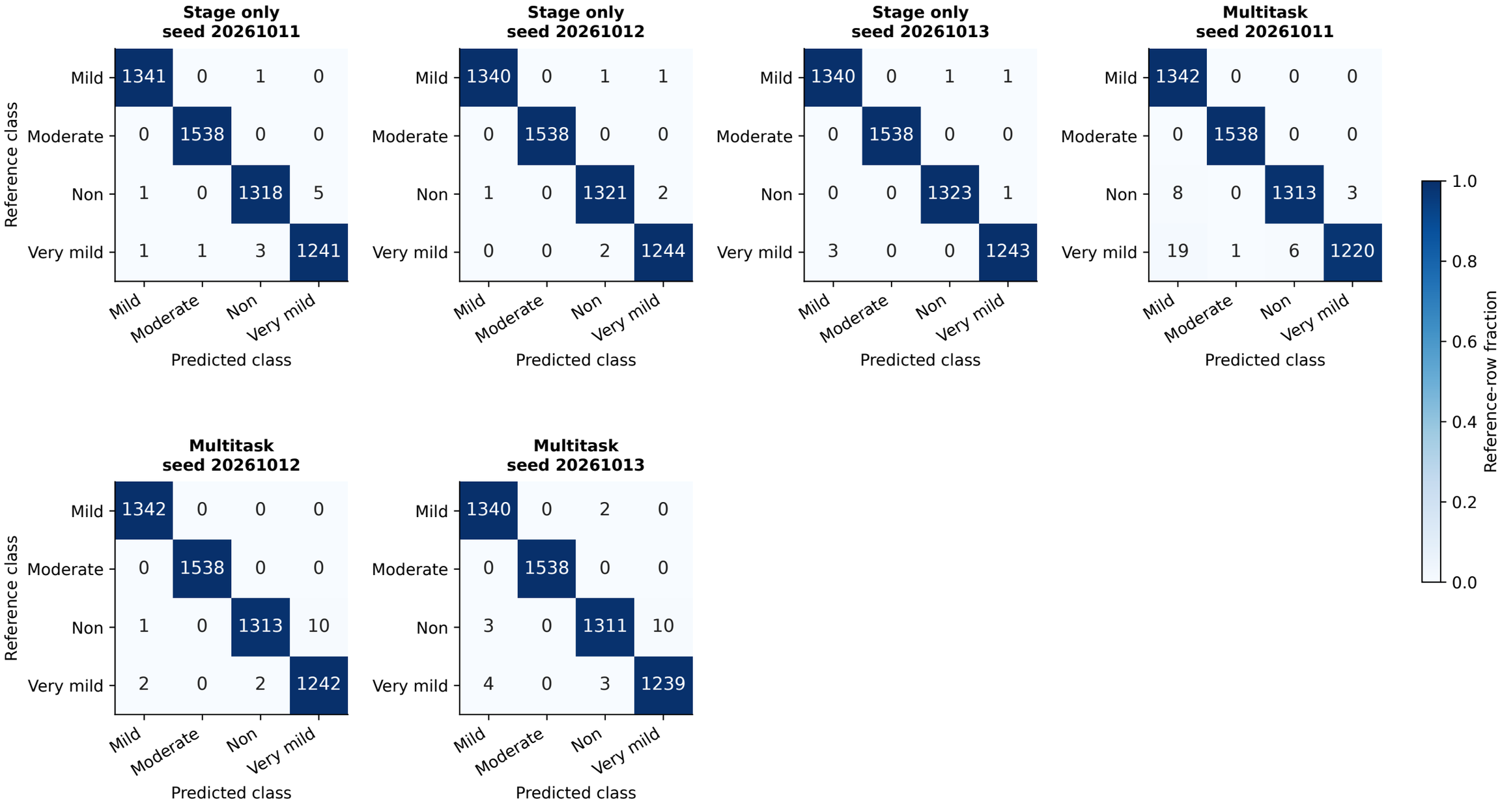
Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.



Stored best.pt validation counts; these development data selected the checkpoints. No additional inference. All six candidates shown; old/ensemble validation matrices unavailable.

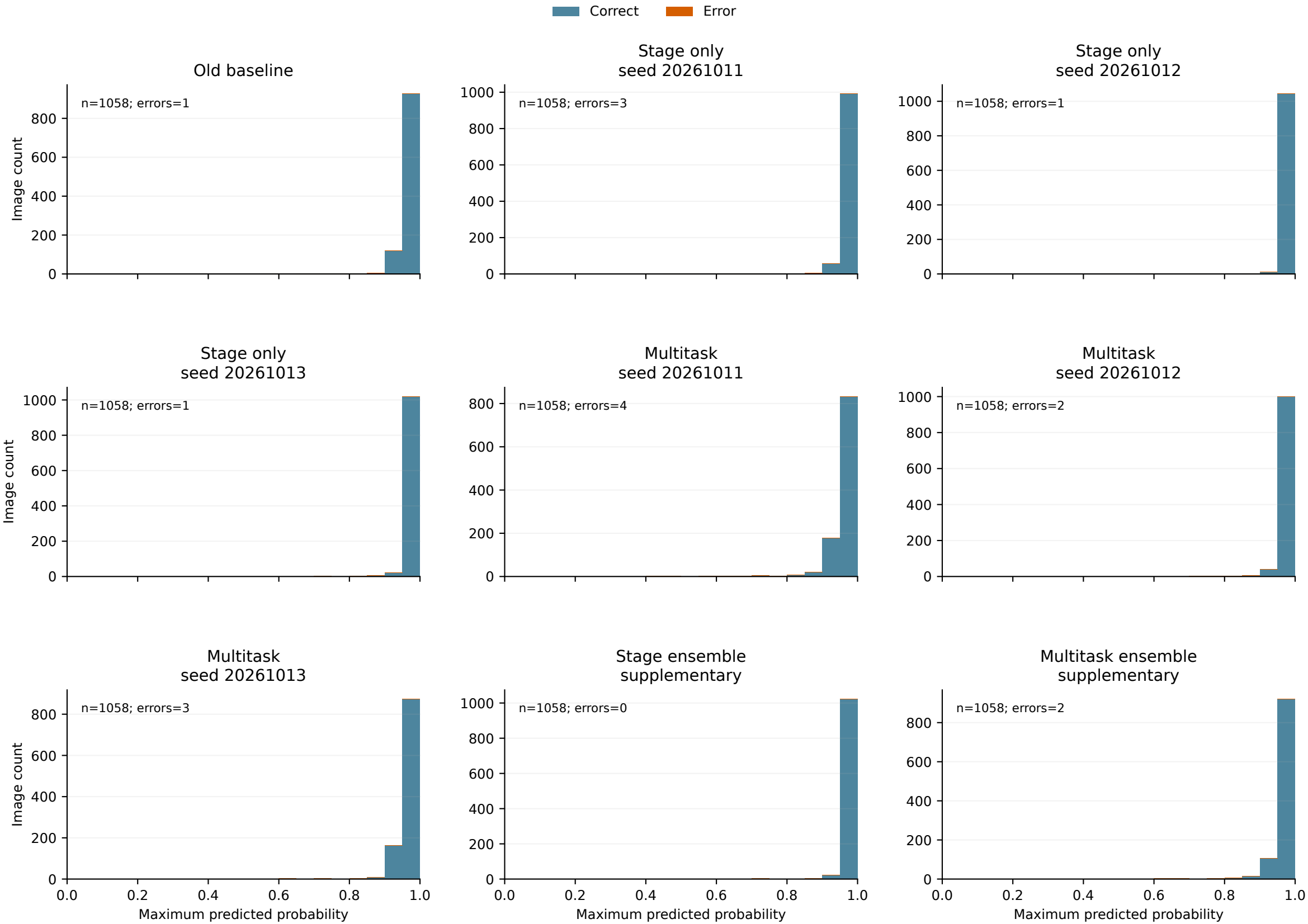
Full new test5450 | additional post-hoc evaluation | n=5450

Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.



Additional clean evaluation of the entire new test split; it does not replace the frozen1058 primary or756 sensitivity. All six frozen candidates; no model selection. Image groups are not patient identities.

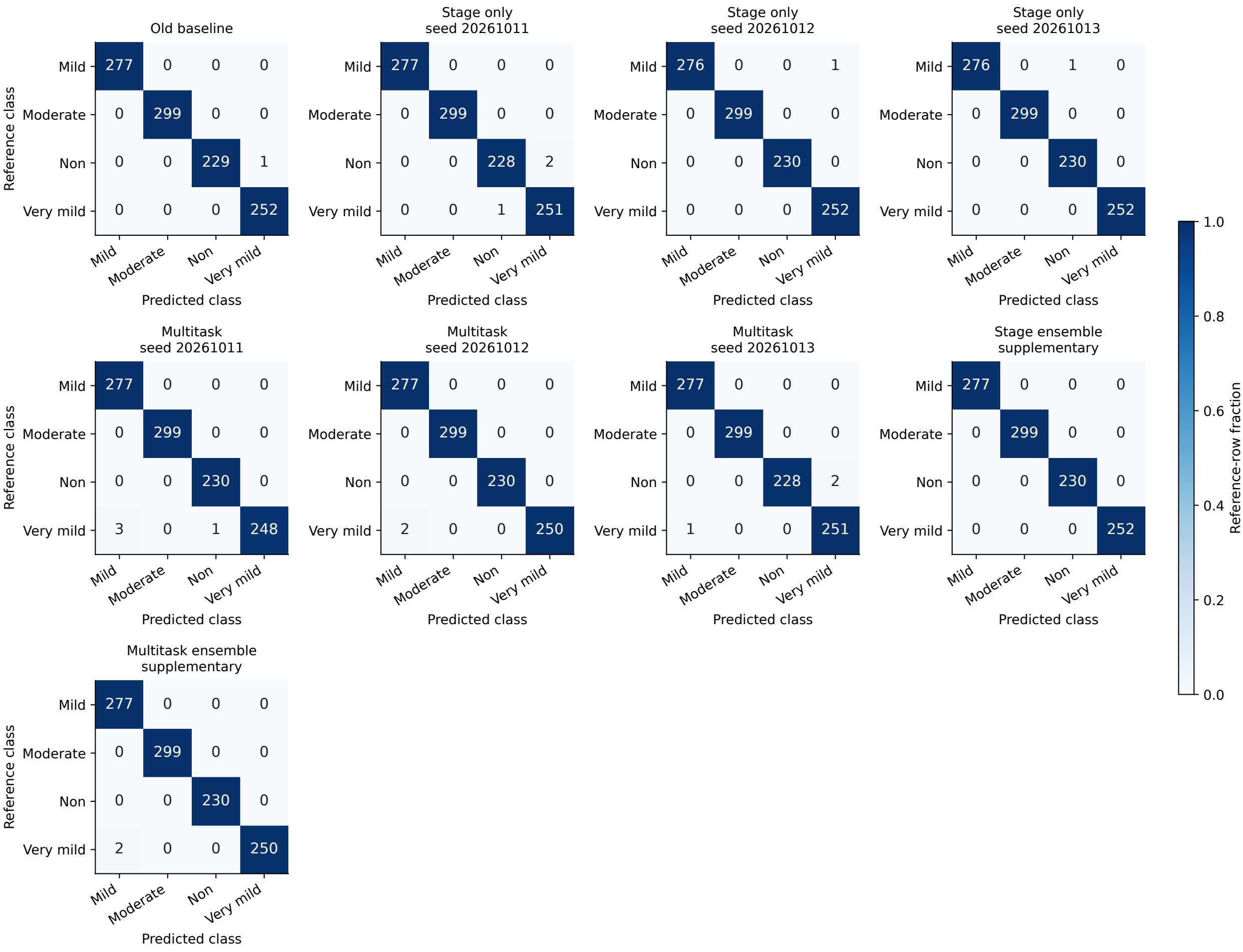
Primary 1,058 images: exact-row held out | confidence distributions



302 images have old development hash-group mates. Hash groups are not patient identities. Fixed 0.05-wide bins; panel count axes vary. Confidence is not validated clinical certainty. Both ensembles are supplementary.

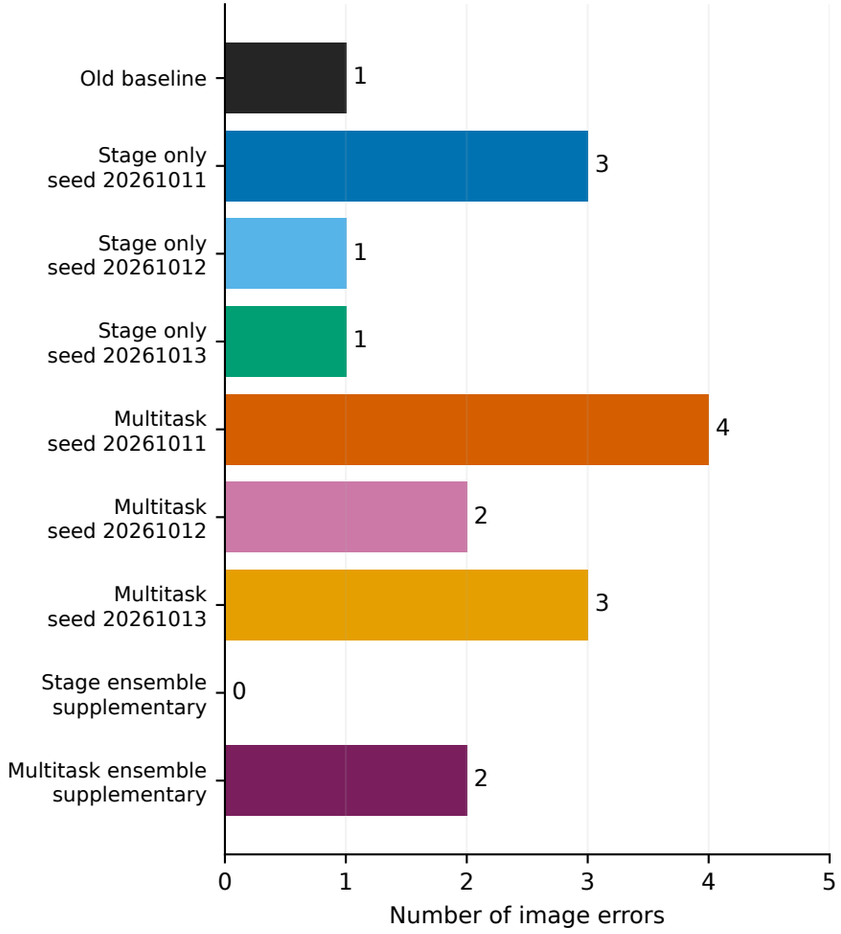
Primary 1,058 images: exact-row held out | n=1058

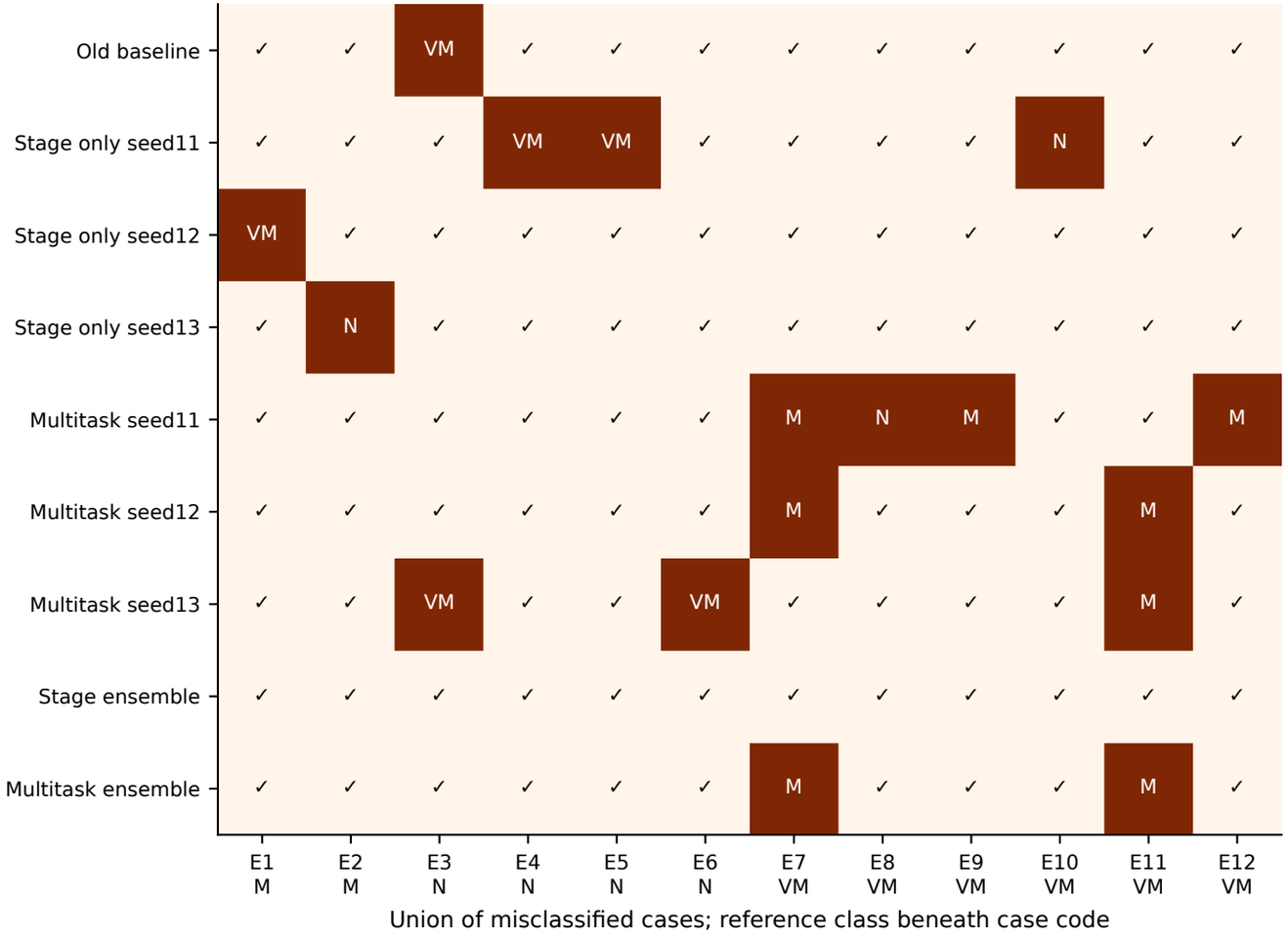
Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.



302 images have old development hash-group mates. Hash groups are not patient identities. Both fixed ensembles are supplementary.

Primary 1,058 images: exact-row held out | error audit

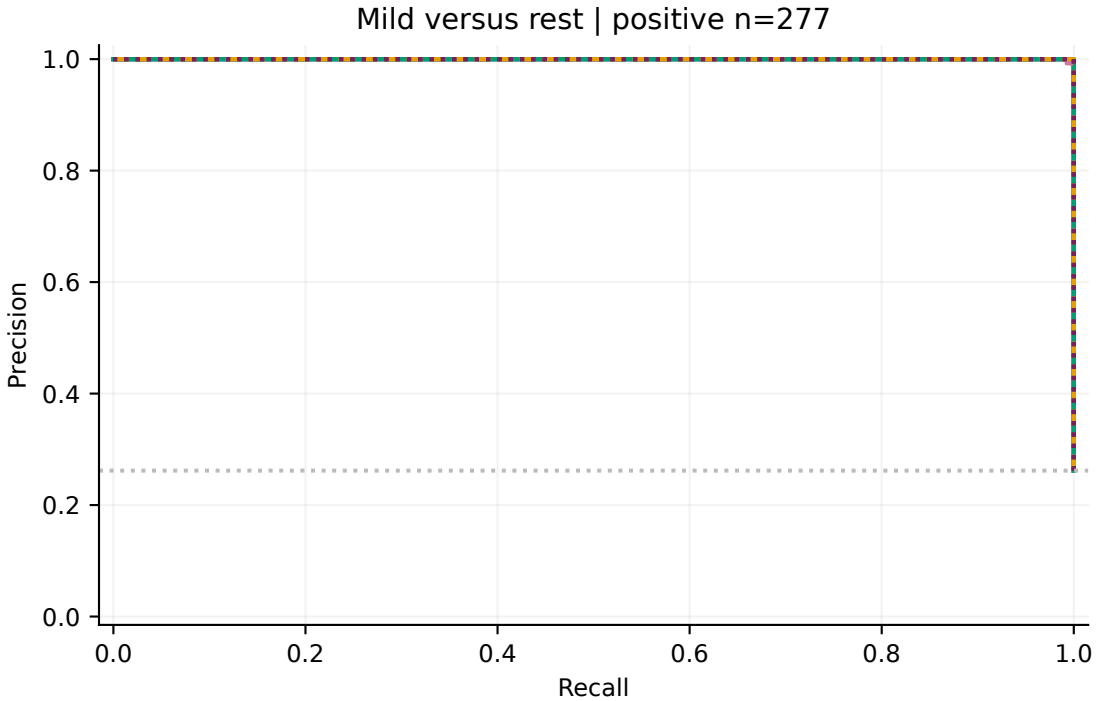


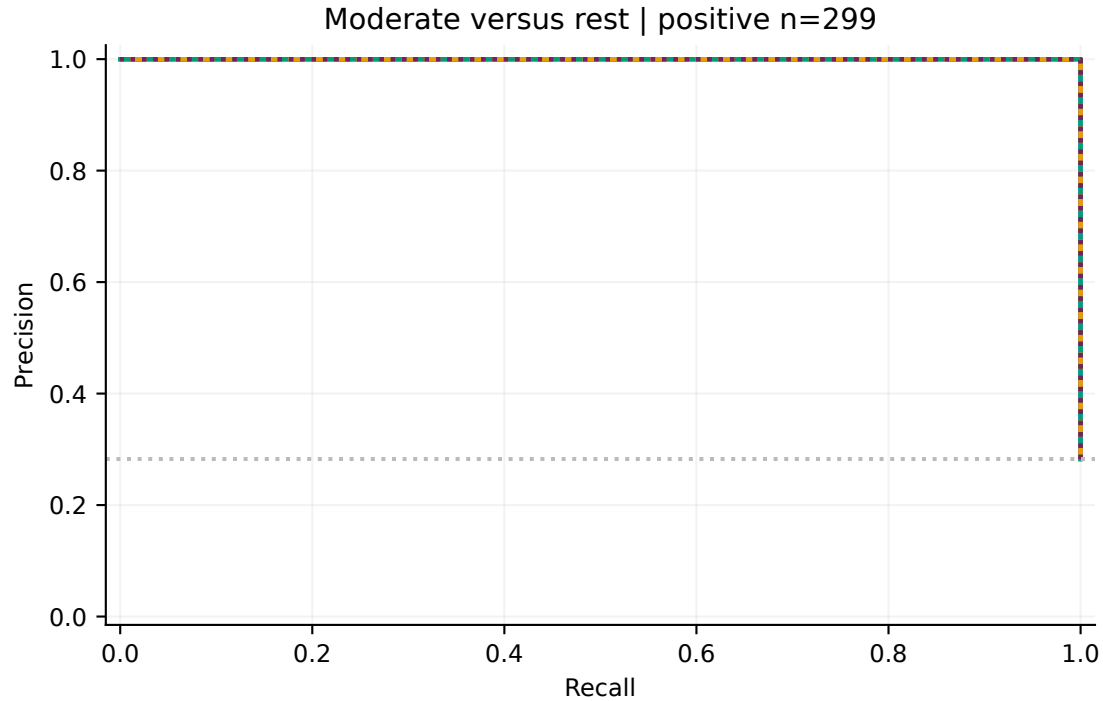


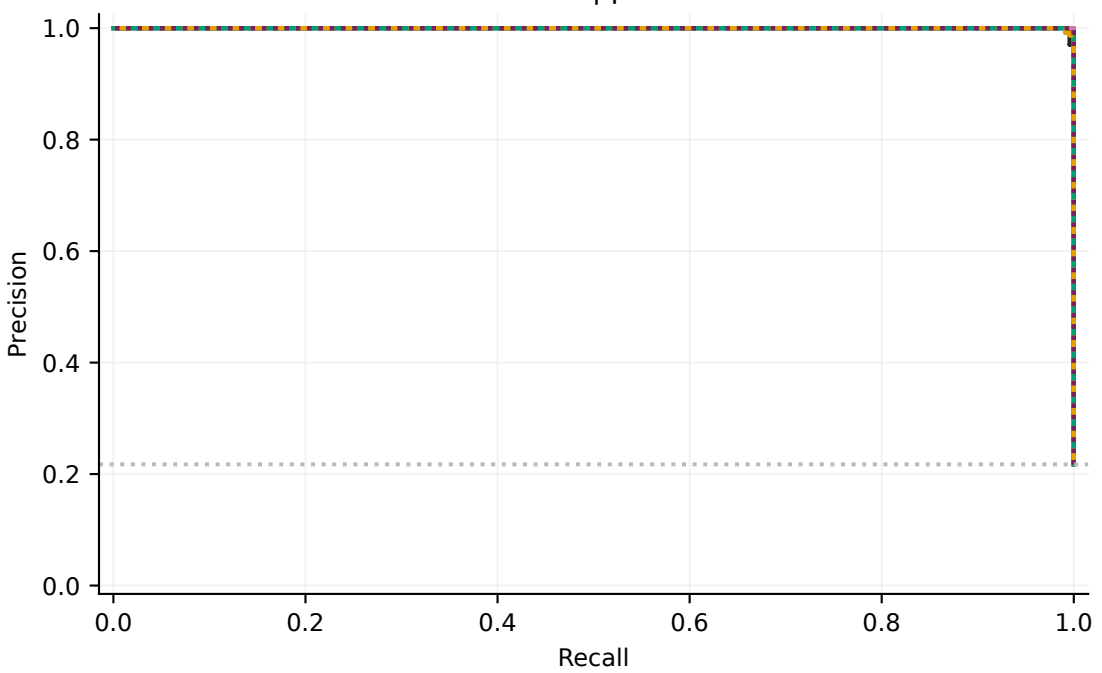
302 images have old development hash-group mates. Hash groups are not patient identities. Every missed case is shown; orange means error. M=Mild, Mod=Moderate, N=Non, VM=Very mild. Case identities: error_cases.csv. Both ensembles are supplementary.

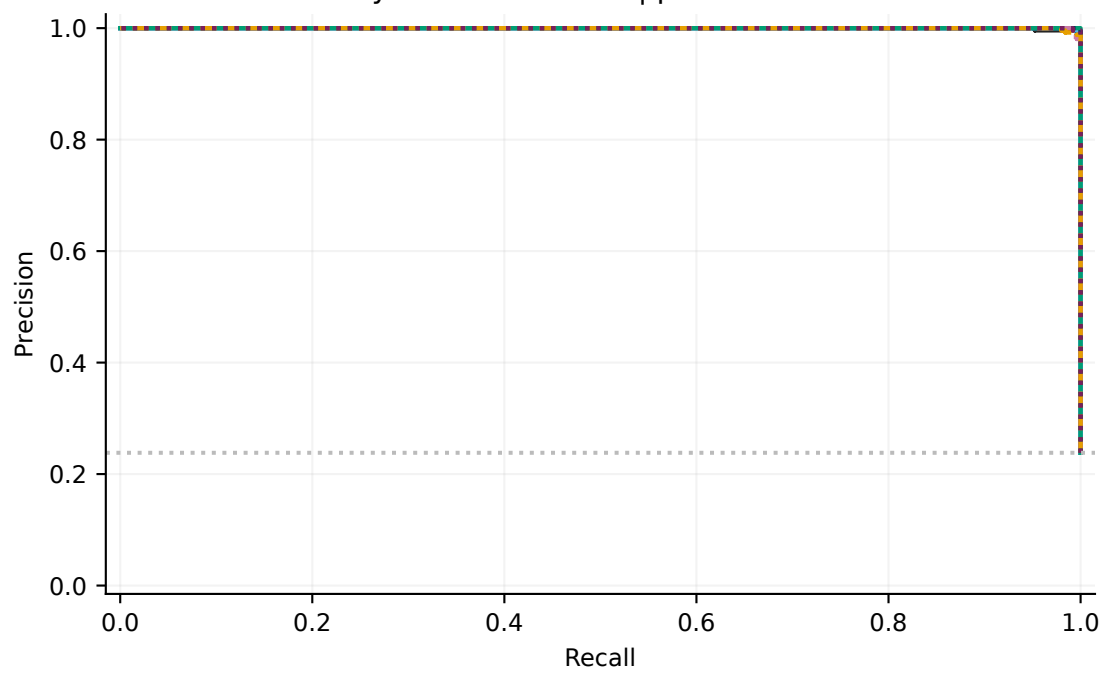
Primary 1,058 images: exact-row held out | precision–recall

Old baseline
Stage only | seed 20261011
Stage only | seed 20261012
Stage only | seed 20261013
Multitask | seed 20261011
Multitask | seed 20261012
Multitask | seed 20261013
Stage ensemble | supplementary
Multitask ensemble | supplementary



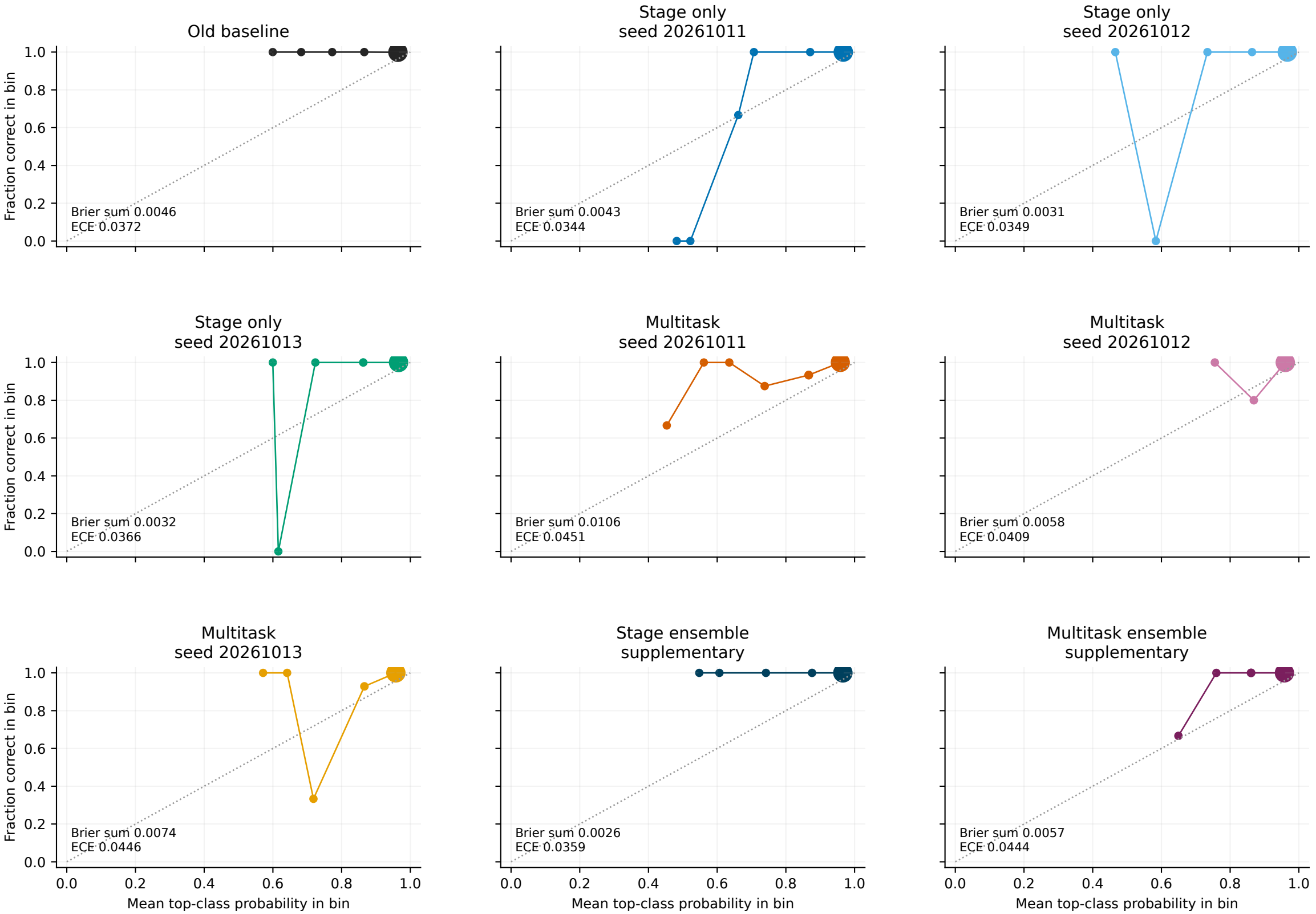






302 images have old development hash-group mates. Hash groups are not patient identities. Fixed model order; overlapping near-ceiling curves remain in the numeric table. Curves use frozen probabilities, with no threshold selection. Ensembles are supplementary.

Primary 1,058 images: exact-row held out | reliability



302 images have old development hash-group mates. Hash groups are not patient identities. Ten fixed equal-width bins; empty bins omitted; marker area reflects count. Descriptive only; no recalibration or fitted thresholds. Ensembles are supplementary.

Primary 1,058 images: exact-row held out | ROC

Old baseline
Stage only | seed 20261011
Stage only | seed 20261012
Stage only | seed 20261013
Multitask | seed 20261011
Multitask | seed 20261012
Multitask | seed 20261013
Stage ensemble | supplementary
Multitask ensemble | supplementary

Mild versus rest | positive n=277

Moderate versus rest | positive n=299

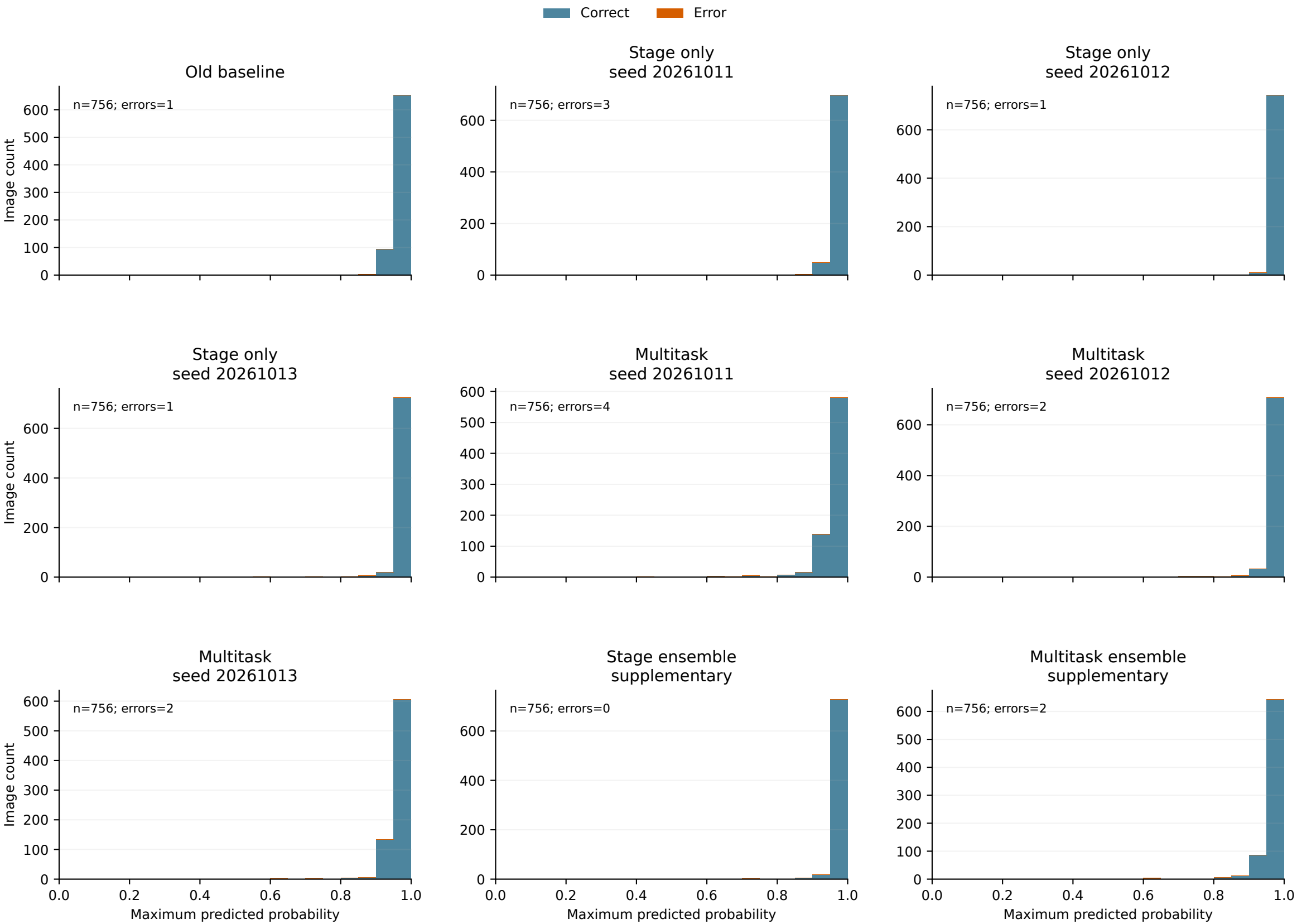
Non versus rest | positive n=230

Very mild versus rest | positive n=252

True-positive rate
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0

302 images have old development hash-group mates. Hash groups are not patient identities. Fixed model order; overlapping near-ceiling curves remain in the numeric table. Curves use frozen probabilities, with no threshold selection. Ensembles are supplementary.

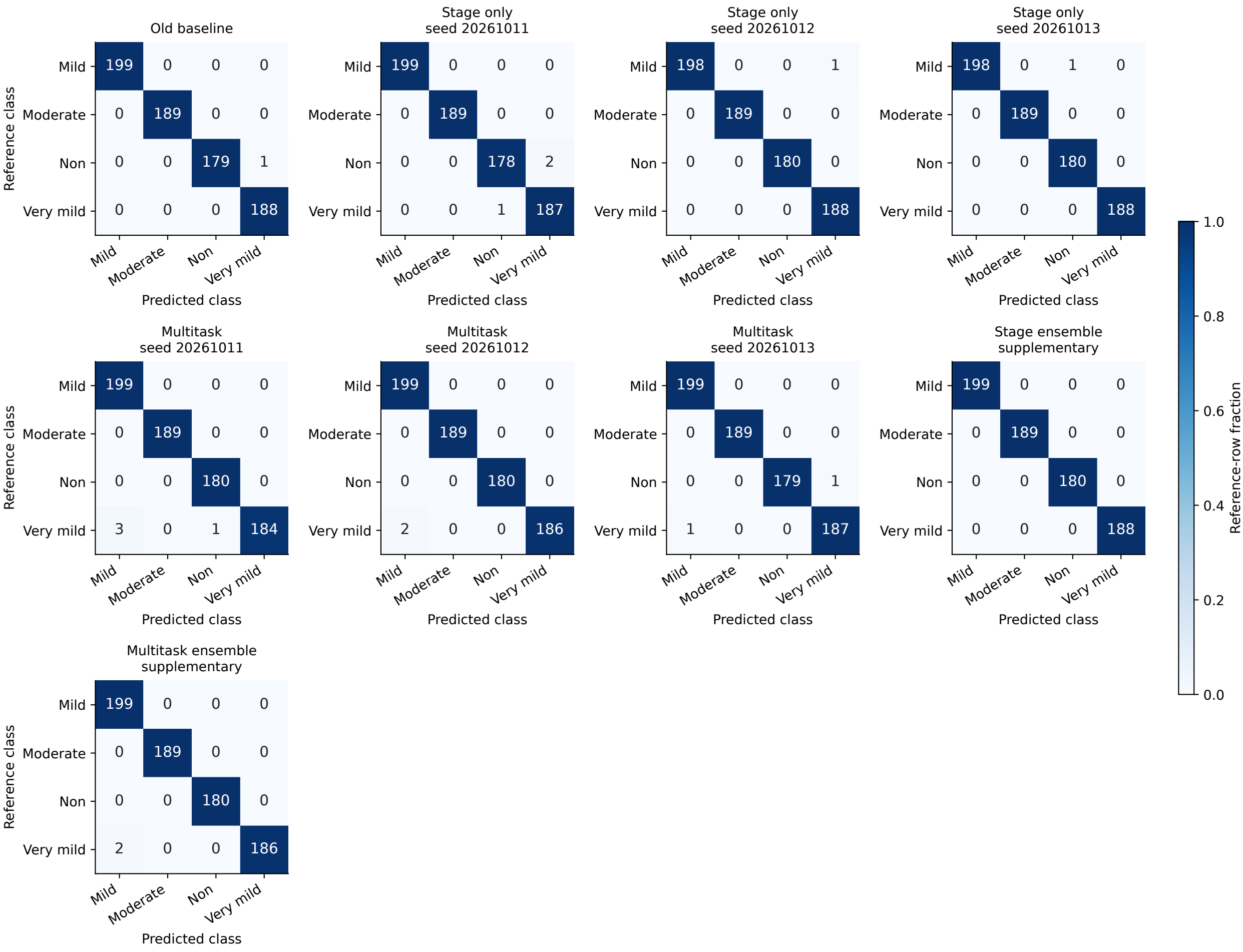
Secondary sensitivity: 756 images | confidence distributions



Excludes observed old development hash-group mates; patient independence remains unknown. Fixed 0.05-wide bins; panel count axes vary. Confidence is not validated clinical certainty. Both ensembles are supplementary.

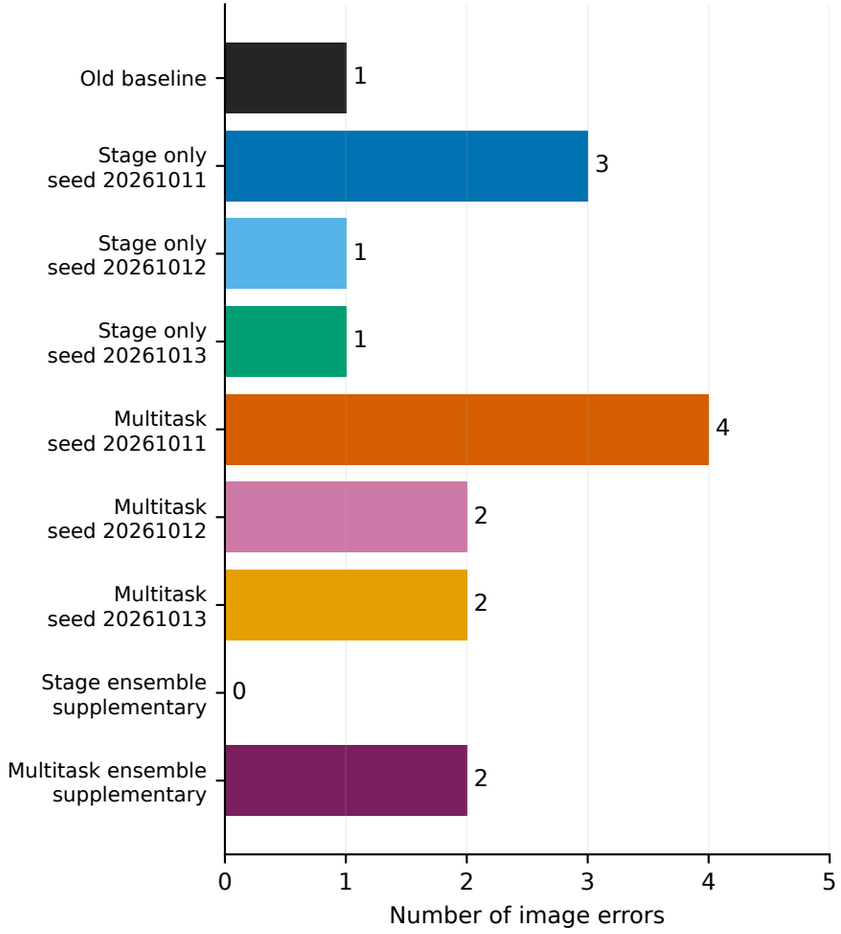
Secondary sensitivity: 756 images | n=756

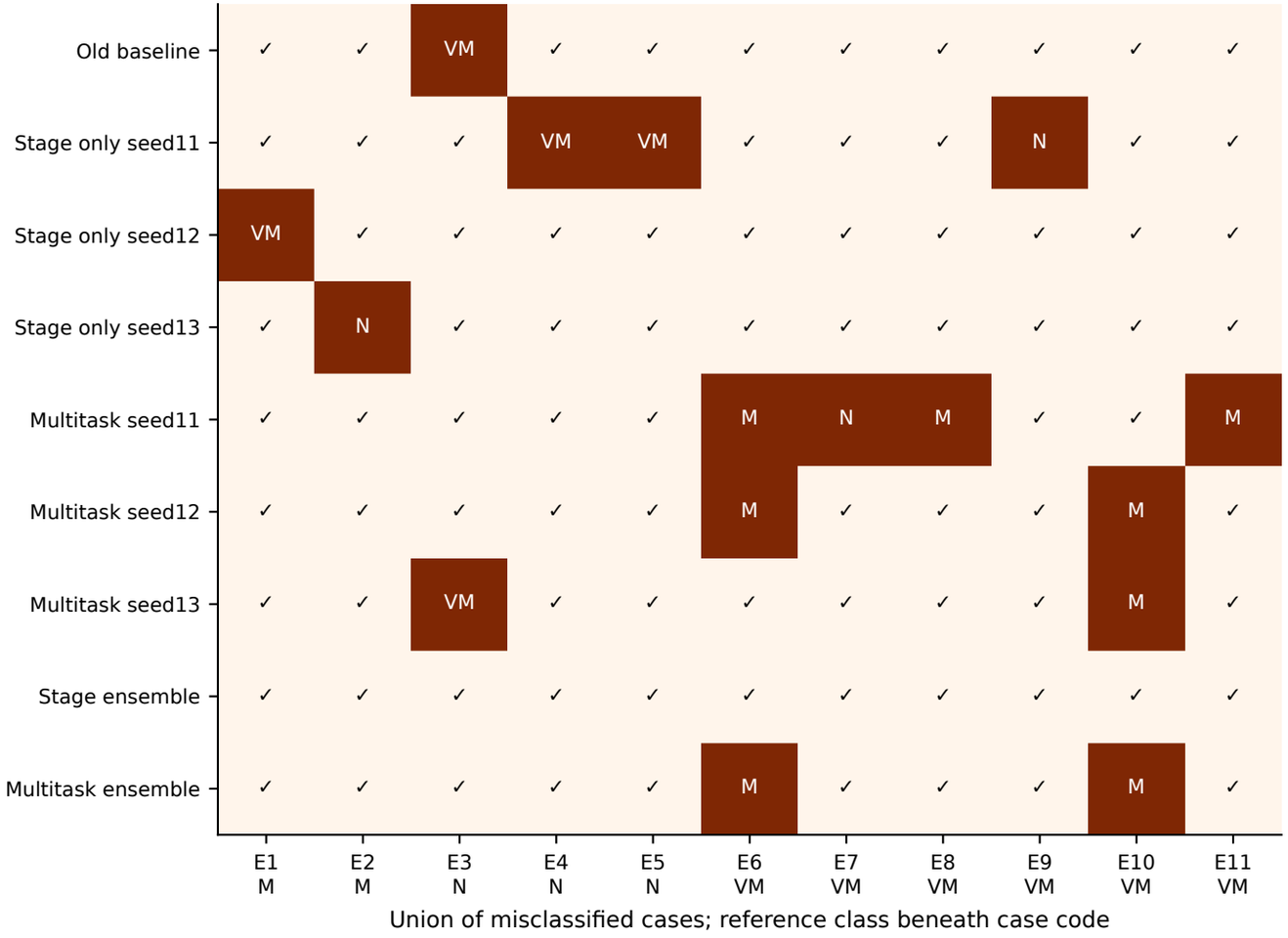
Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.



Excludes observed old development hash-group mates; patient independence remains unknown. Both fixed ensembles are supplementary.

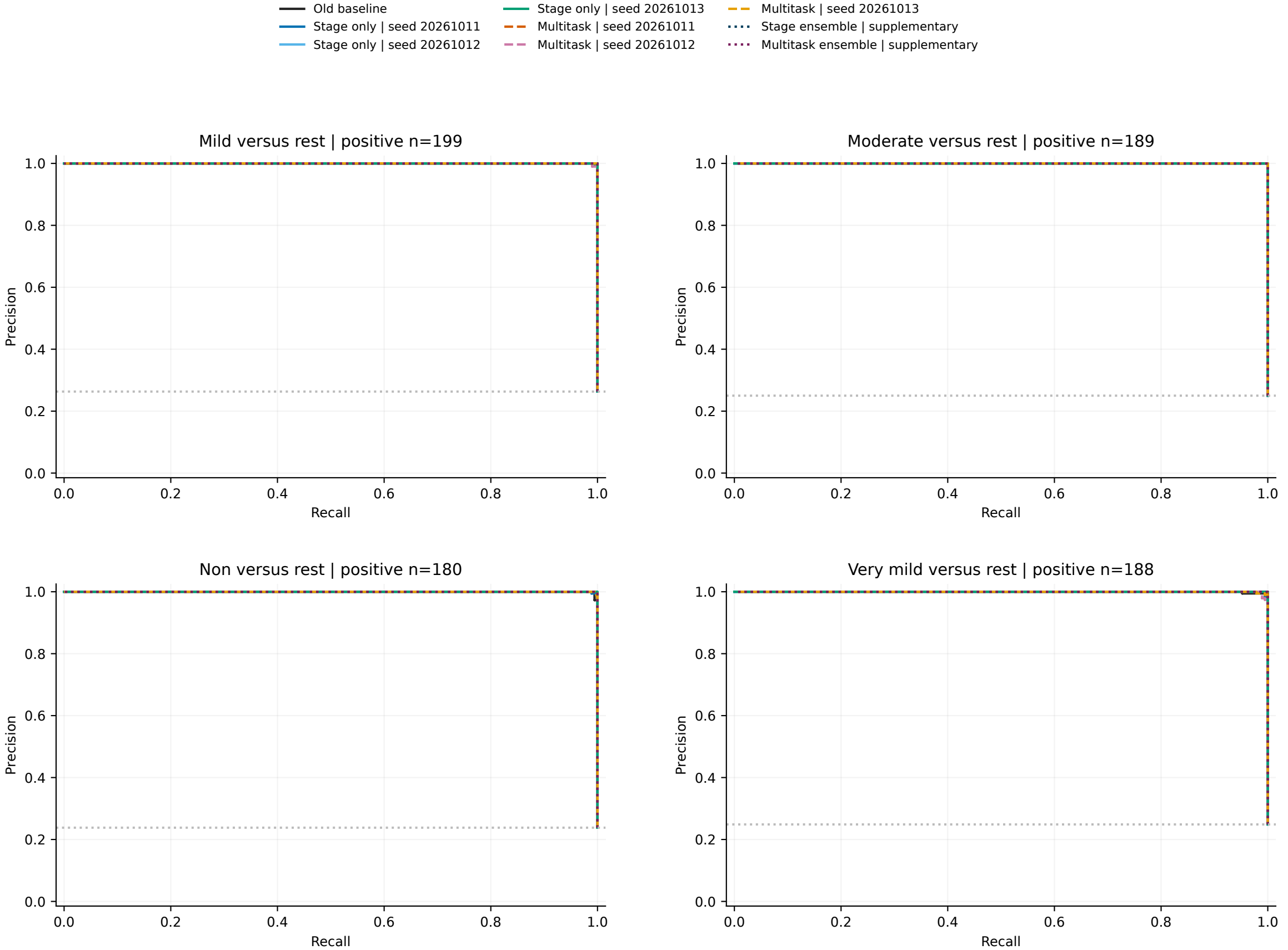
Secondary sensitivity: 756 images | error audit





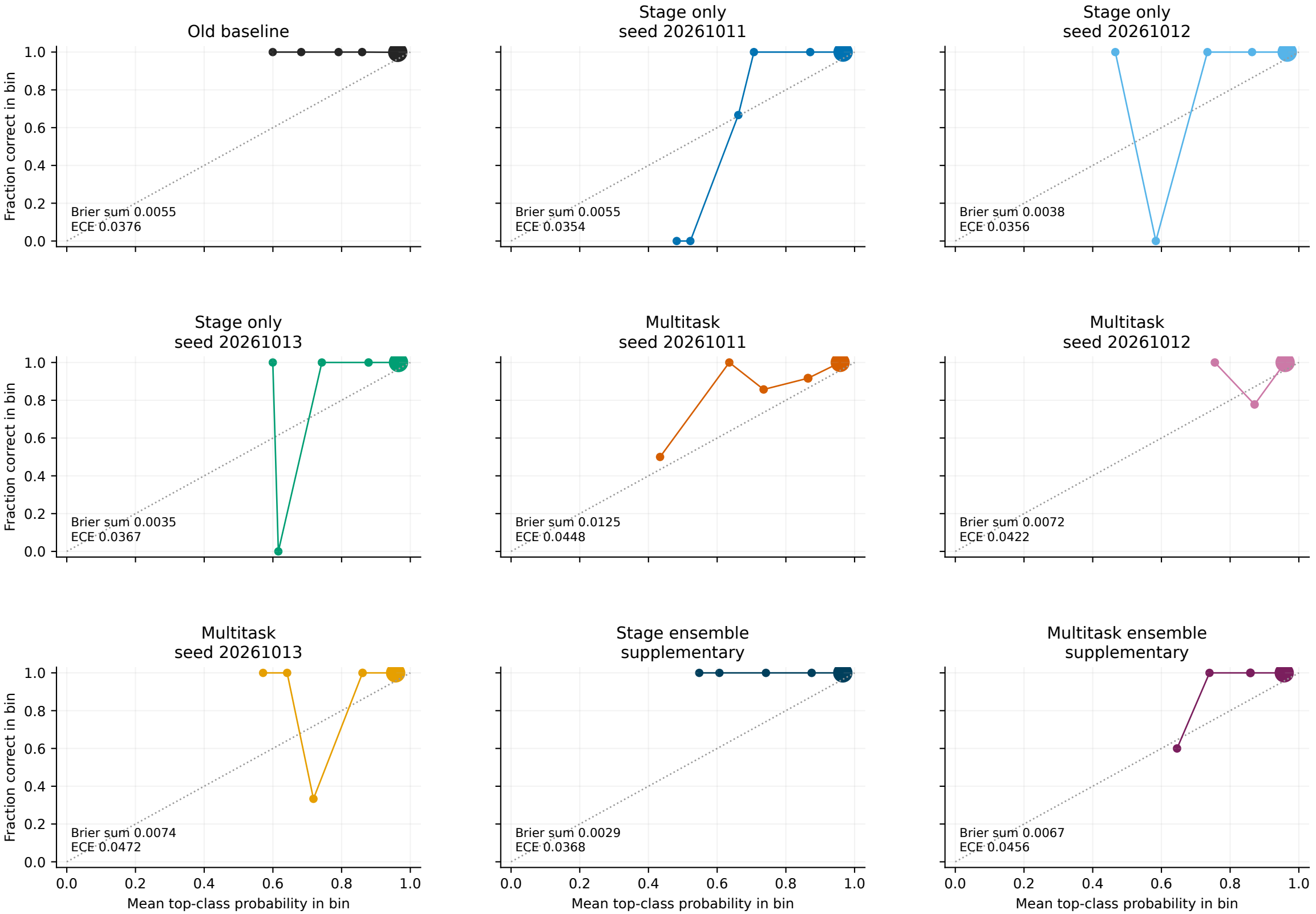
Excludes observed old development hash-group mates; patient independence remains unknown. Every missed case is shown; orange means error. M=Mild, Mod=Moderate, N=Non, VM=Very mild. Case identities: error_cases.csv. Both ensembles are supplementary.

Secondary sensitivity: 756 images | precision–recall



Excludes observed old development hash-group mates; patient independence remains unknown. Fixed model order; overlapping near-ceiling curves remain in the numeric table. Curves use frozen probabilities, with no threshold selection. Ensembles are supplementary.

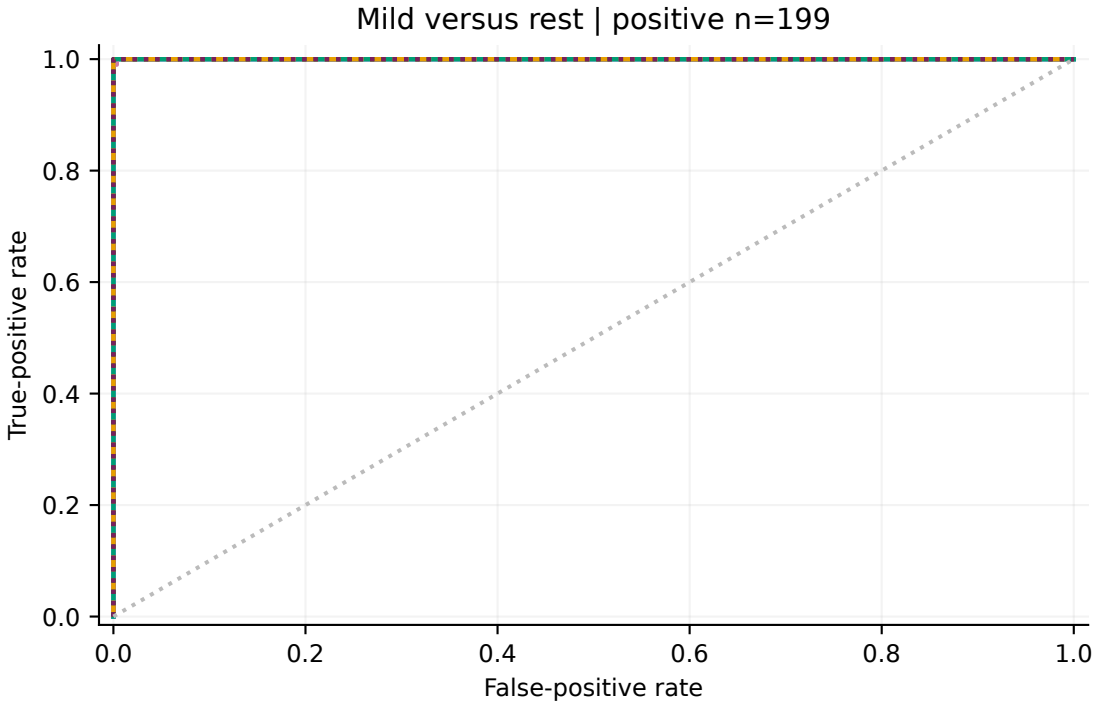
Secondary sensitivity: 756 images | reliability

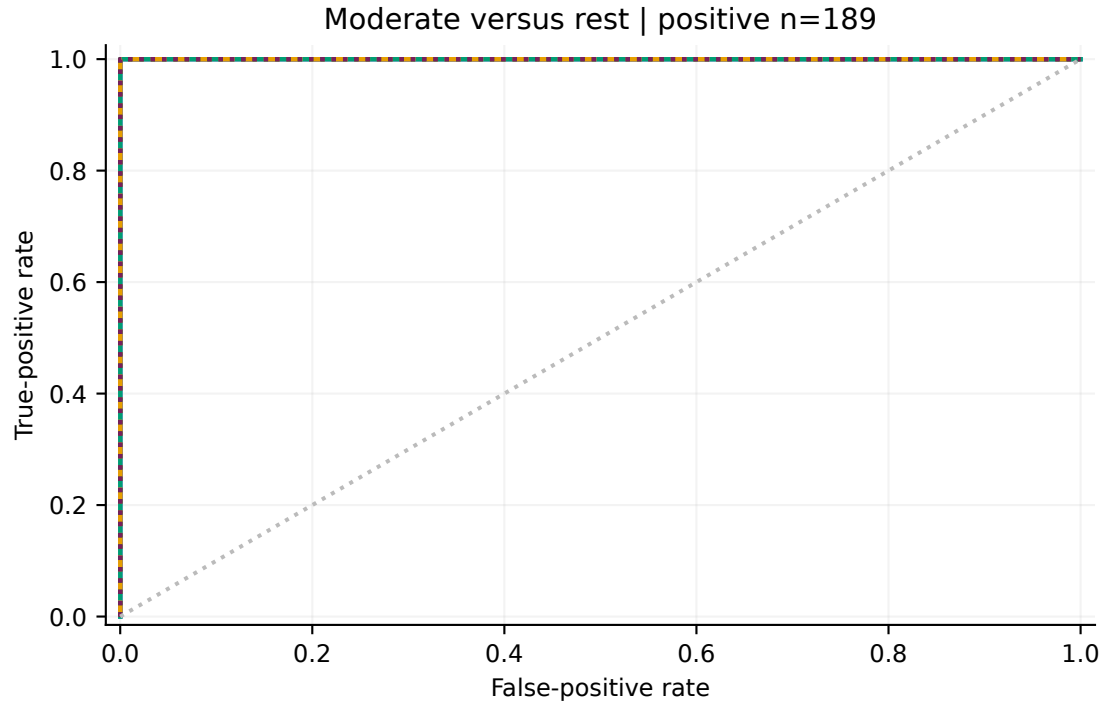


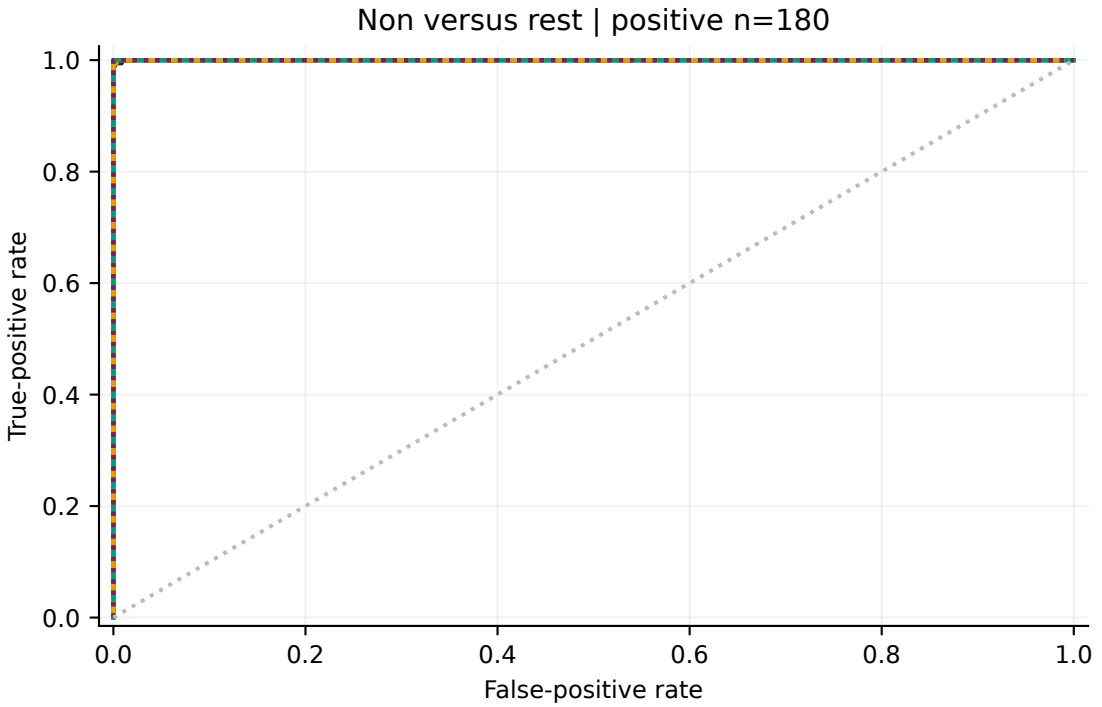
Excludes observed old development hash-group mates; patient independence remains unknown. Ten fixed equal-width bins; empty bins omitted; marker area reflects count. Descriptive only; no recalibration or fitted thresholds. Ensembles are supplementary.

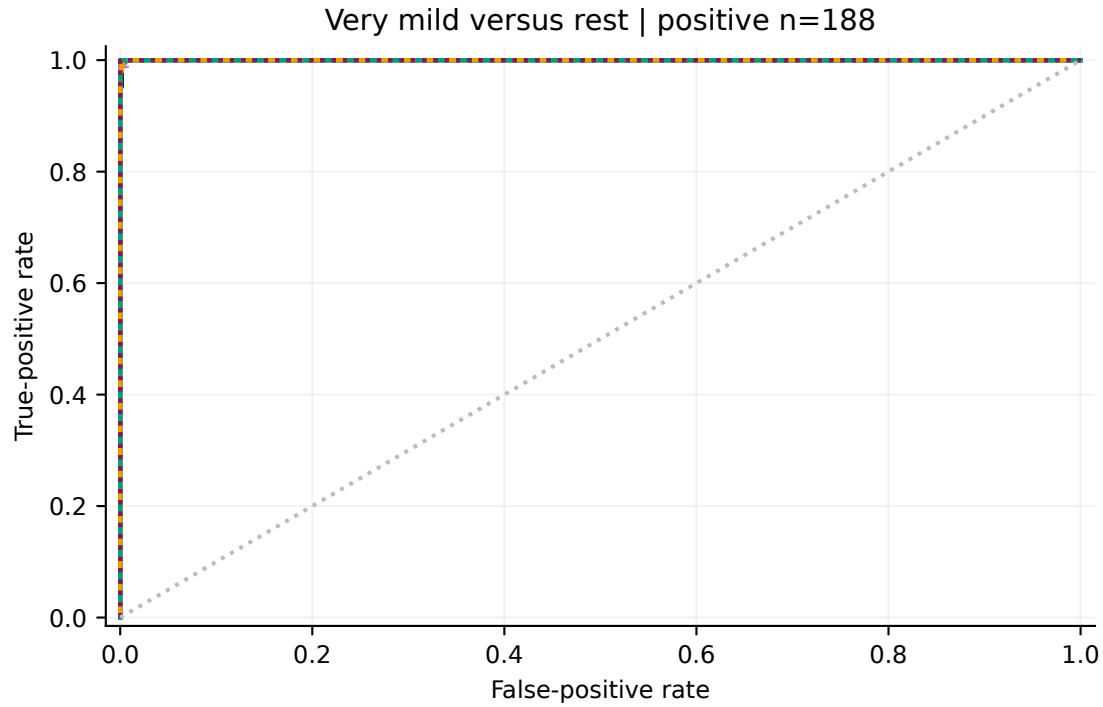
Secondary sensitivity: 756 images | ROC

Old baseline
Stage only | seed 20261011
Stage only | seed 20261012
Stage only | seed 20261013
Multitask | seed 20261011
Multitask | seed 20261012
Multitask | seed 20261013
Stage ensemble | supplementary
Multitask ensemble | supplementary



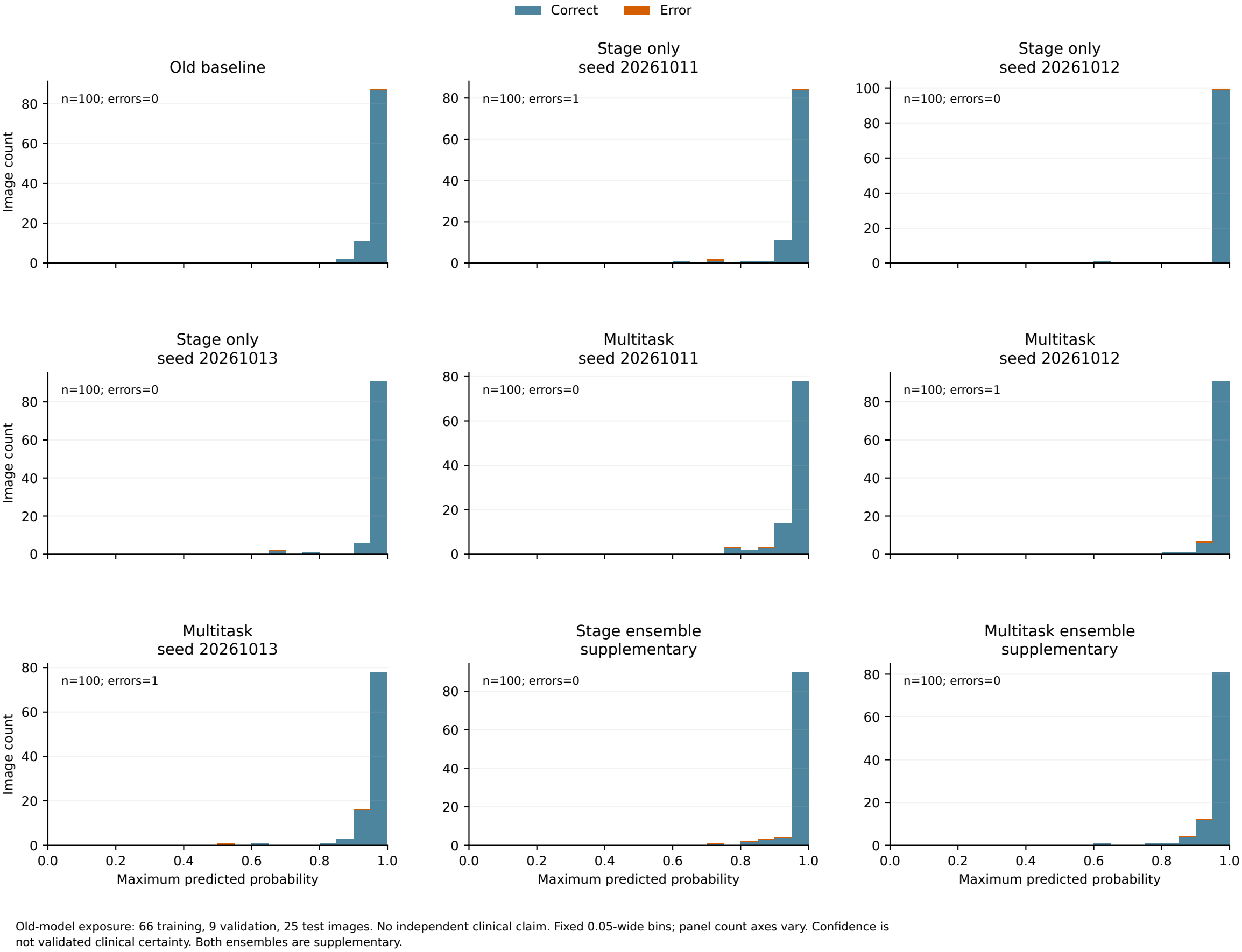






Excludes observed old development hash-group mates; patient independence remains unknown. Fixed model order; overlapping near-ceiling curves remain in the numeric table. Curves use frozen probabilities, with no threshold selection. Ensembles are supplementary.

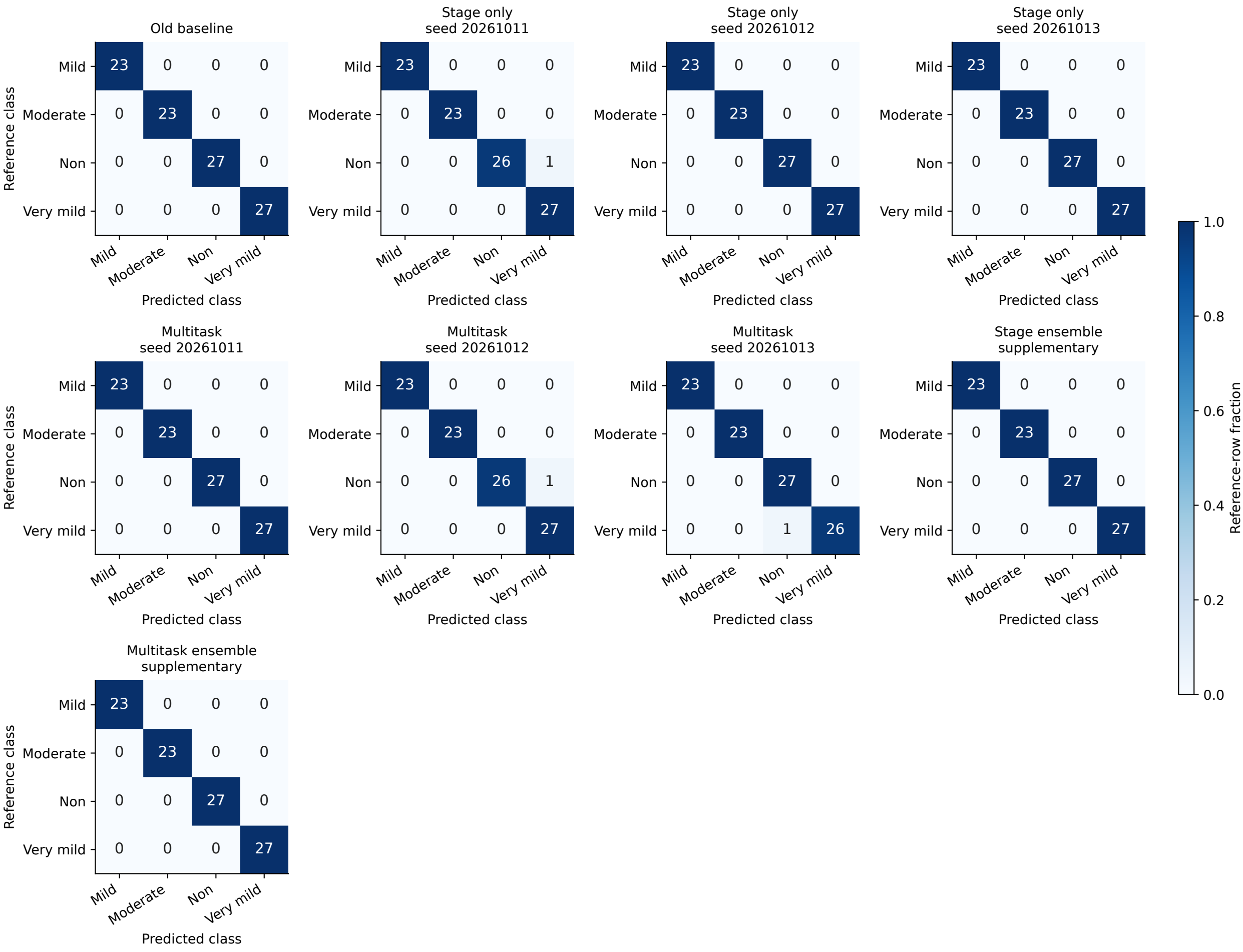
Original 100: historically reused image benchmark | confidence distributions



Old-model exposure: 66 training, 9 validation, 25 test images. No independent clinical claim. Fixed 0.05-wide bins; panel count axes vary. Confidence is not validated clinical certainty. Both ensembles are supplementary.

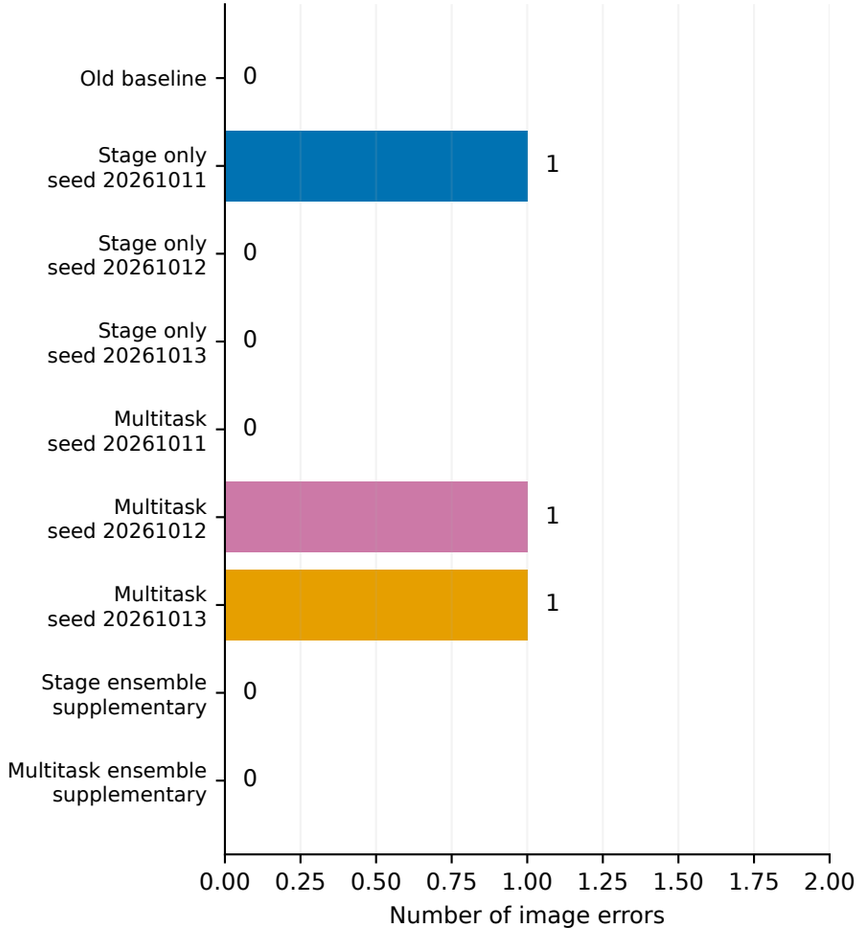
Original 100: historically reused image benchmark | n=100

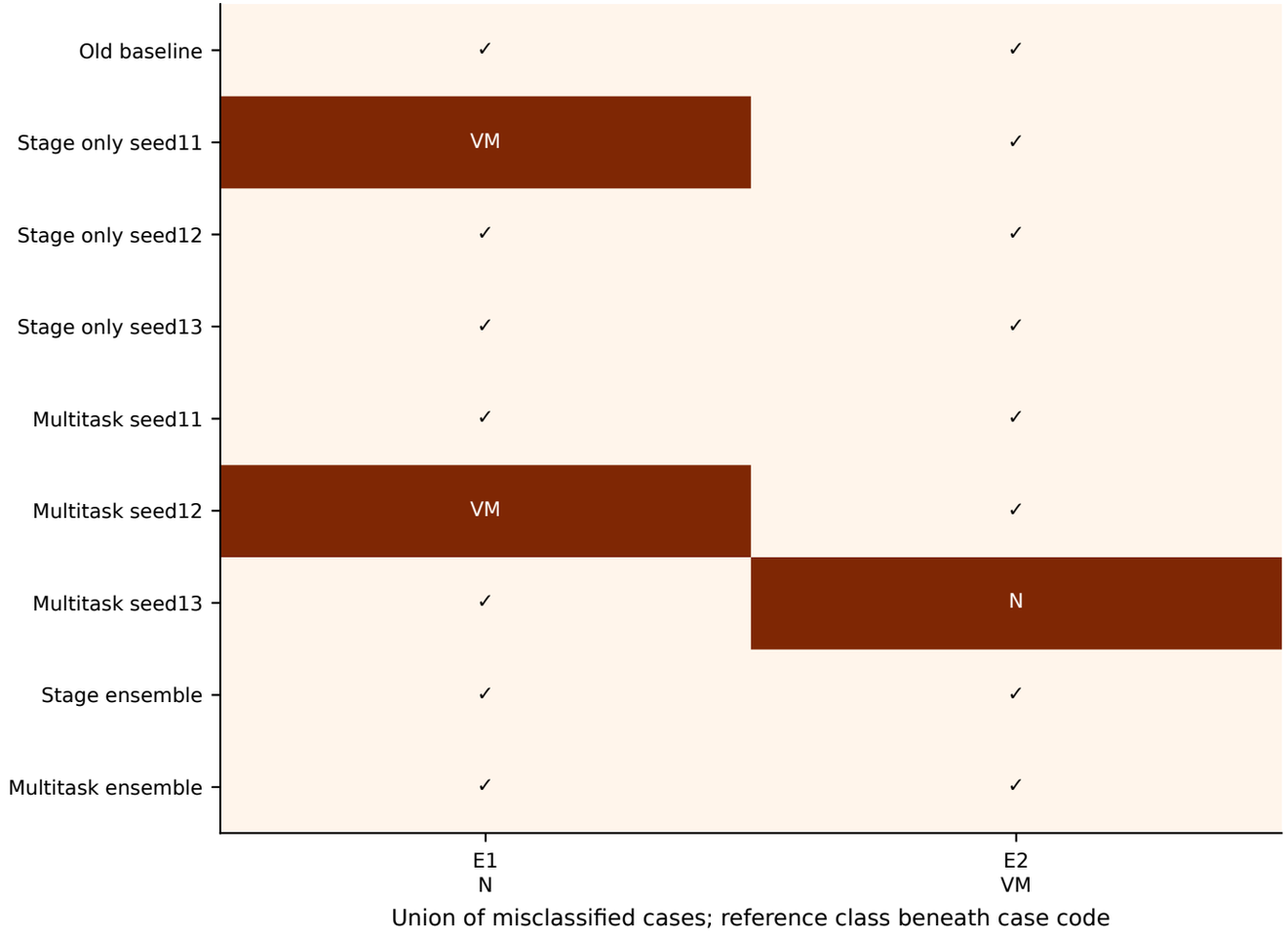
Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.



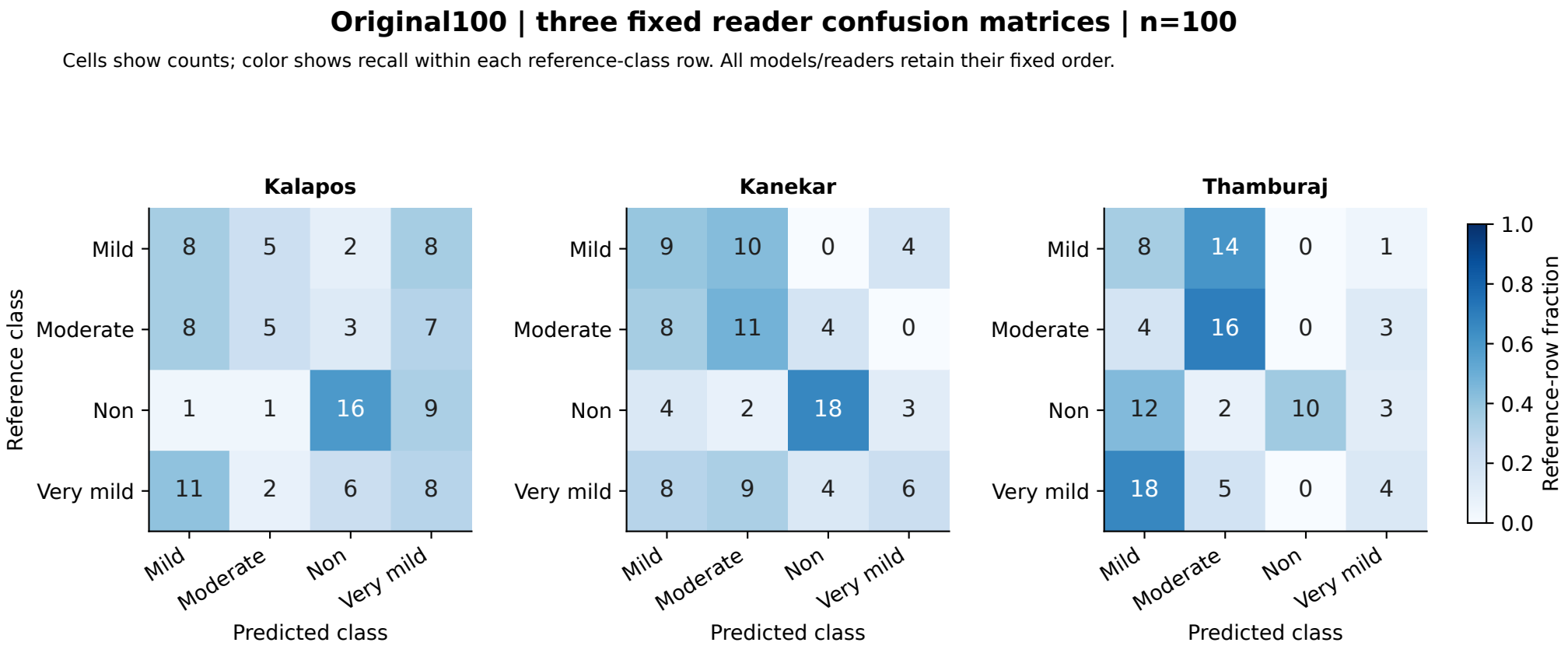
Old-model exposure: 66 training, 9 validation, 25 test images. No independent clinical claim. Both fixed ensembles are supplementary.

Original 100: historically reused image benchmark | error audit

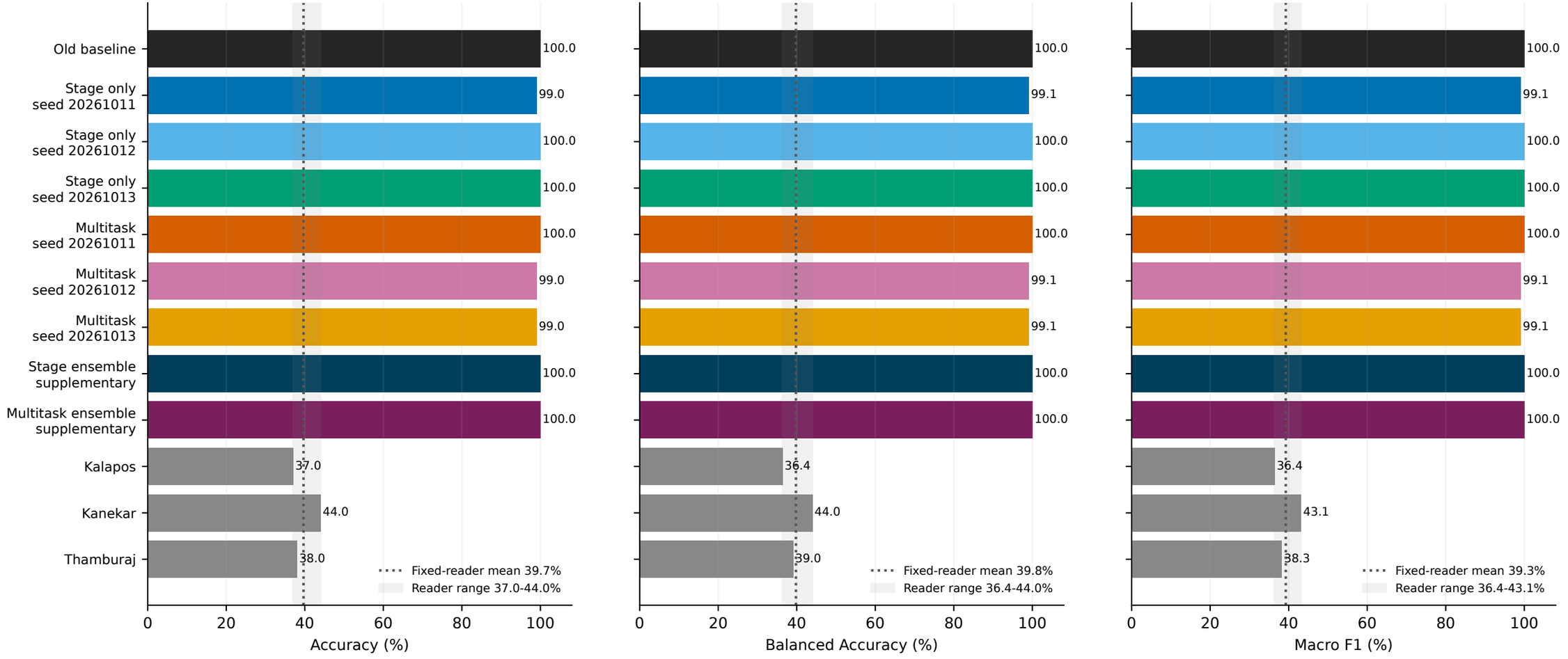




Old-model exposure: 66 training, 9 validation, 25 test images. No independent clinical claim. Every missed case is shown; orange means error. M=Mild, Mod=Moderate, N=Non, VM=Very mild. Case identities: error_cases.csv. Both ensembles are supplementary.

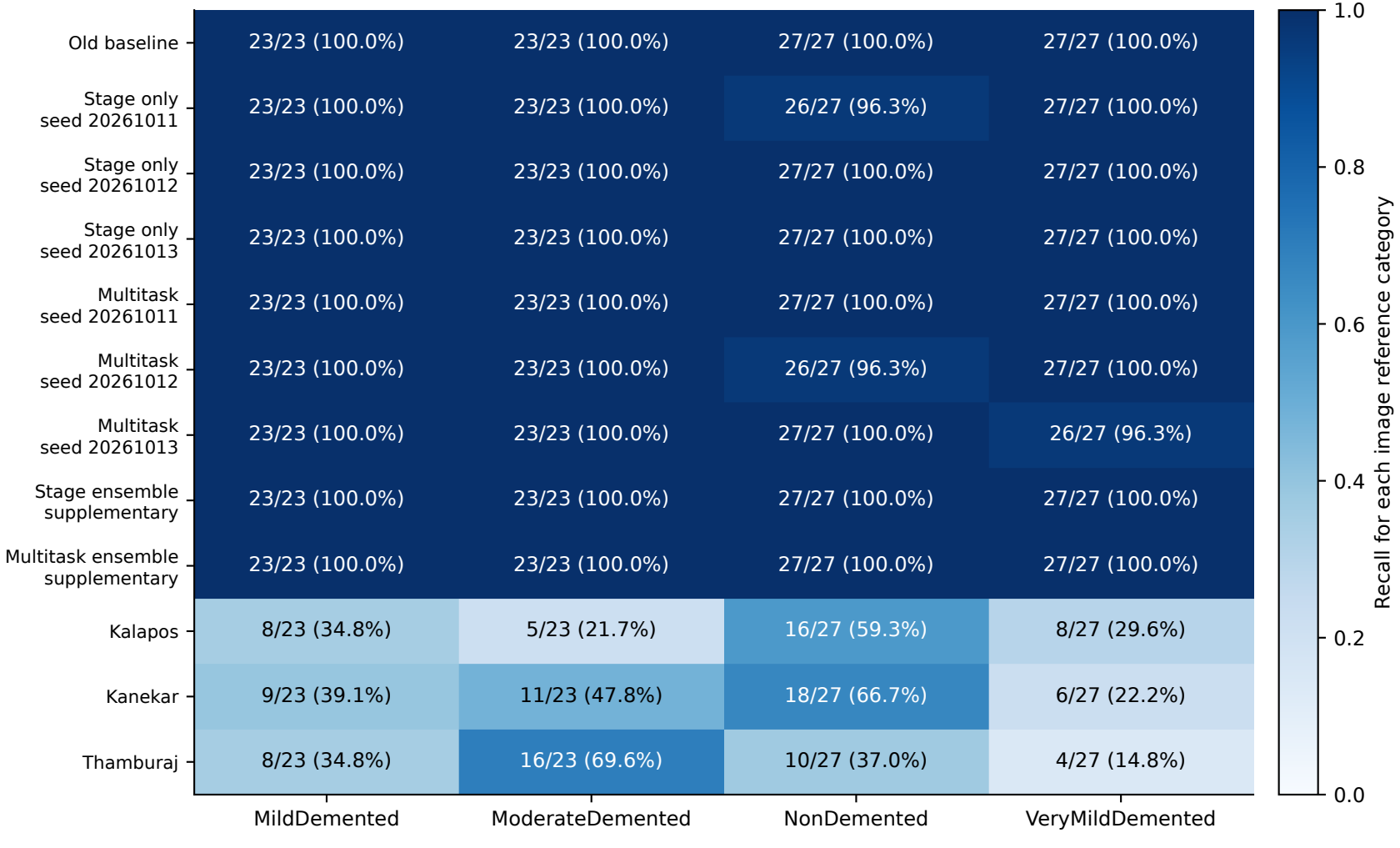

Original100 | three fixed reader confusion matrices | n=100
Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.
Kalapos
Kanekar
Thamburaj
Reference class
Mild
Moderate
Non
Very mild
8 5 2 8
8 5 3 7
1 1 16 9
11 2 6 8
9 10 0 4
8 11 4 0
4 2 18 3
8 9 4 6
8 14 0 1
4 16 0 3
12 2 10 3
18 5 0 4
Predicted class
Reference-row fraction
1.0
0.8
0.6
0.4
0.2
0.0
Old-model exposure: 66 training, 9 validation, 25 test images. No independent clinical claim.

Original100 | all models and three fixed readers



Old-model exposure: 66 training, 9 validation, 25 test images. No independent clinical claim. Dotted lines and shaded bands show means and min-max ranges across these three readers, not confidence intervals. Both ensembles are supplementary. Reader-case rows are not pooled as independent observations.

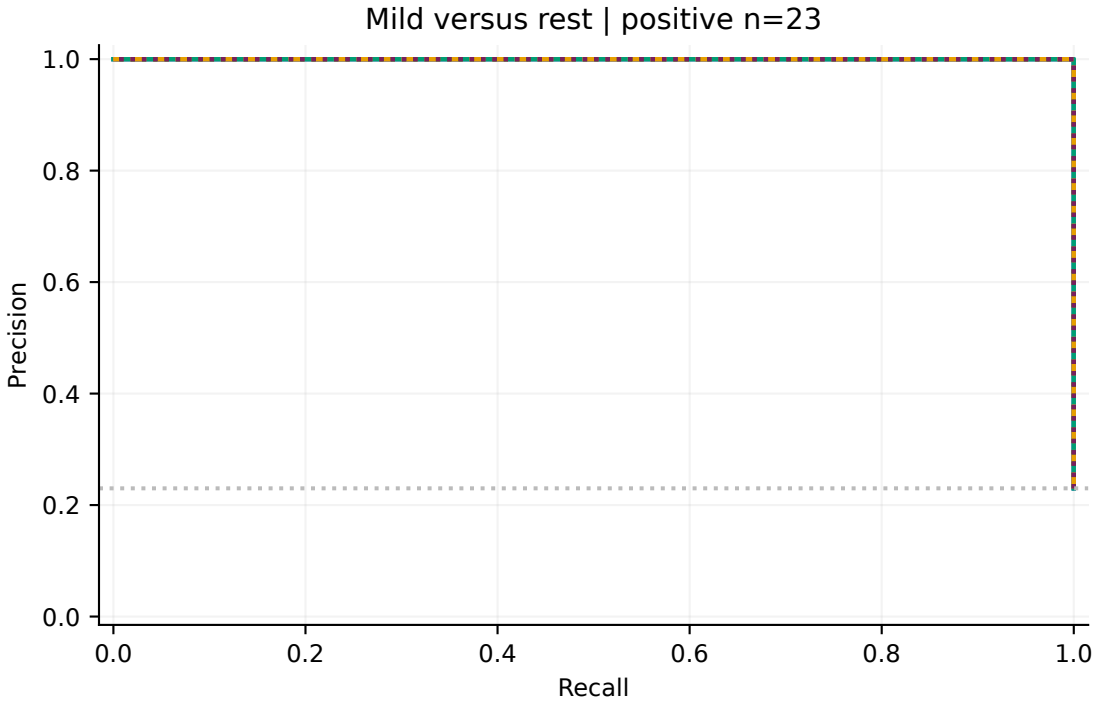
Original100 | per-class image recall

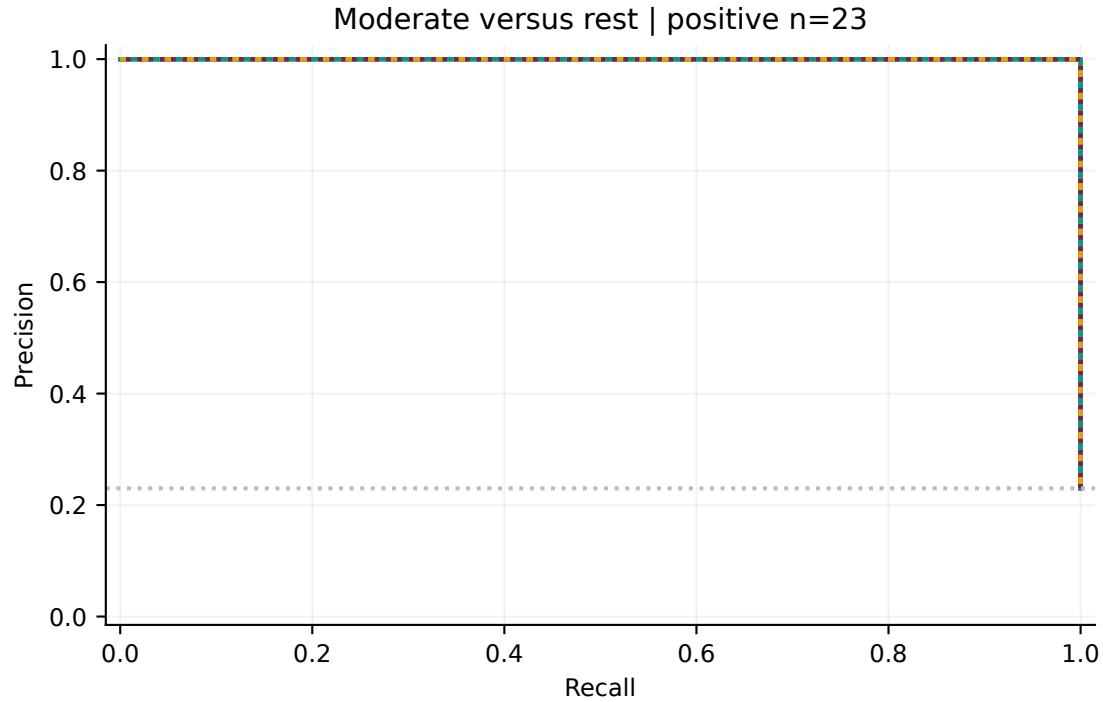


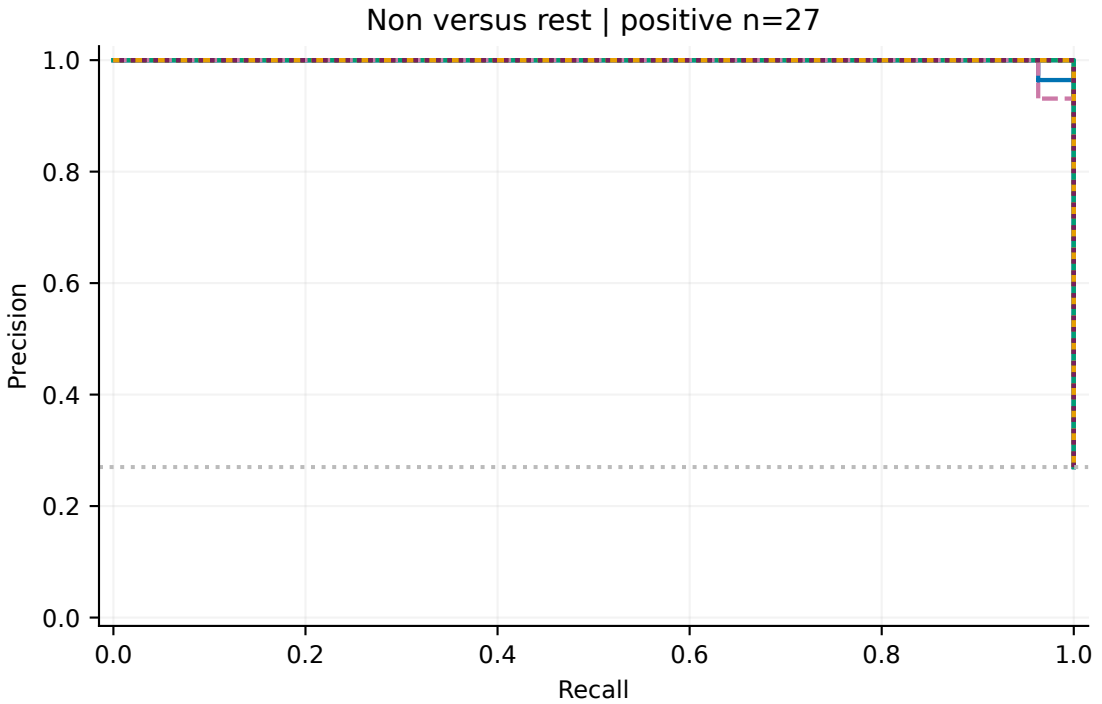
Old-model exposure: 66 training, 9 validation, 25 test images. No independent clinical claim. These legacy image categories are not a patient-level clinical staging reference. Counts preserve class denominators; ensembles are supplementary.

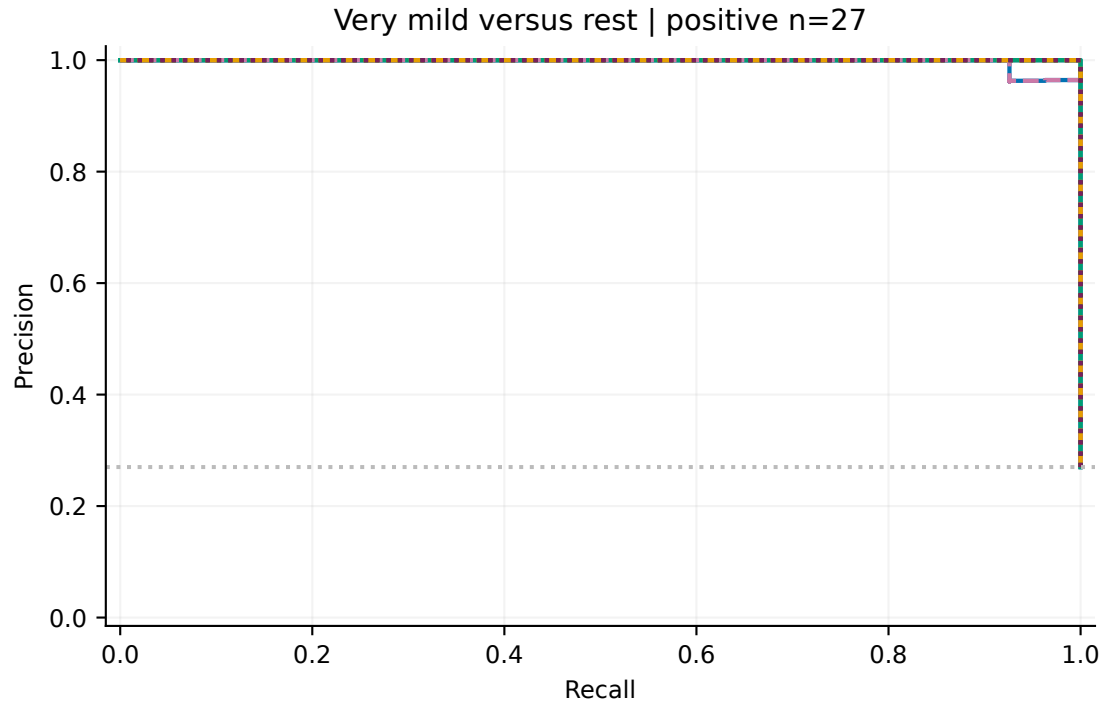
Original 100: historically reused image benchmark | precision–recall

Old baseline
Stage only | seed 20261011
Stage only | seed 20261012
Stage only | seed 20261013
Multitask | seed 20261011
Multitask | seed 20261012
Multitask | seed 20261013
Stage ensemble | supplementary
Multitask ensemble | supplementary



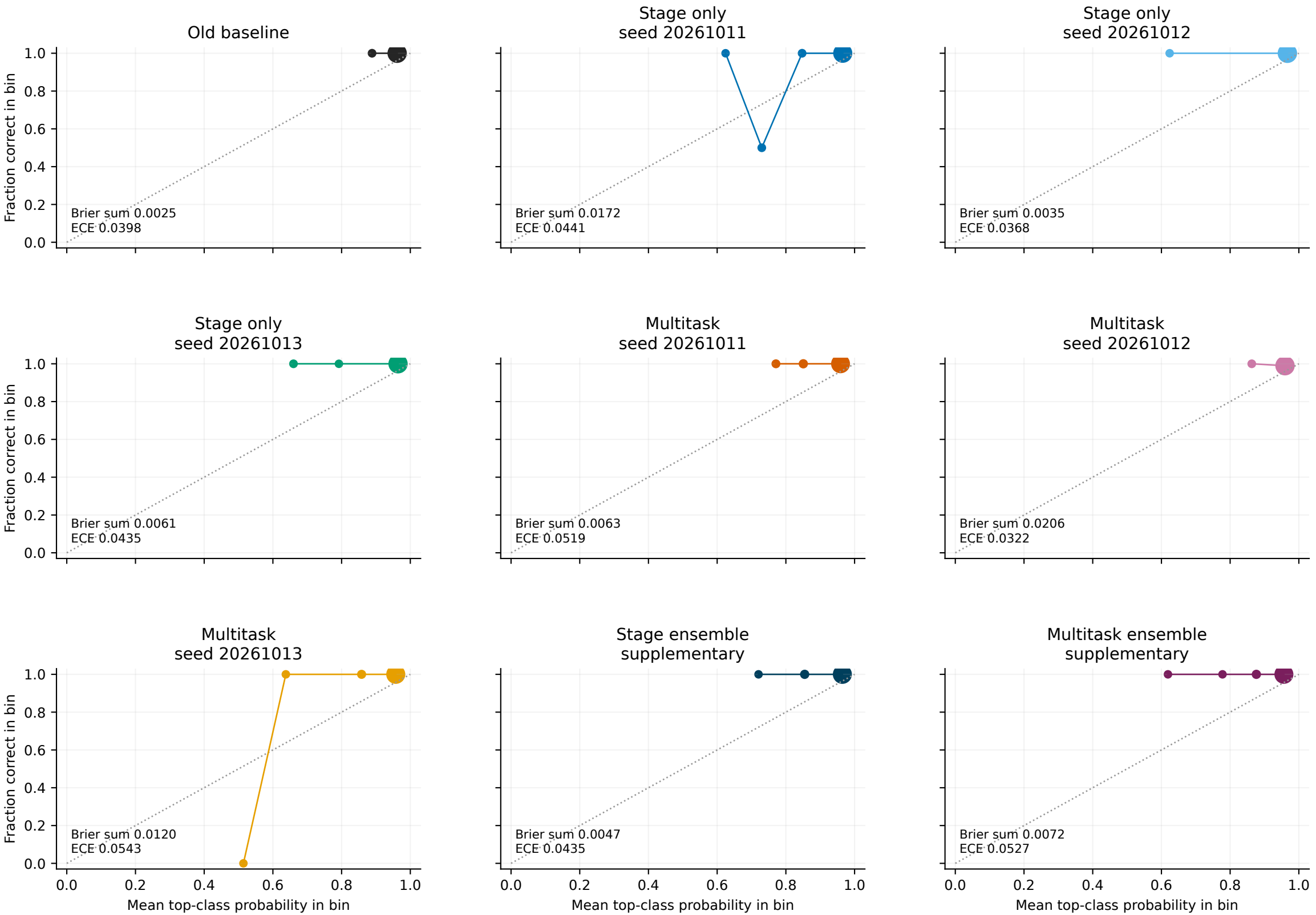






Old-model exposure: 66 training, 9 validation, 25 test images. No independent clinical claim. Fixed model order; overlapping near-ceiling curves remain in the numeric table. Curves use frozen probabilities, with no threshold selection. Ensembles are supplementary.

Original 100: historically reused image benchmark | reliability



Old-model exposure: 66 training, 9 validation, 25 test images. No independent clinical claim. Ten fixed equal-width bins; empty bins omitted; marker area reflects count. Descriptive only; no recalibration or fitted thresholds. Ensembles are supplementary.

Original 100: historically reused image benchmark | ROC

Old baseline
Stage only | seed 20261011
Stage only | seed 20261012
Stage only | seed 20261013
Multitask | seed 20261011
Multitask | seed 20261012
Multitask | seed 20261013
Stage ensemble | supplementary
Multitask ensemble | supplementary

Mild versus rest | positive n=23

Moderate versus rest | positive n=23

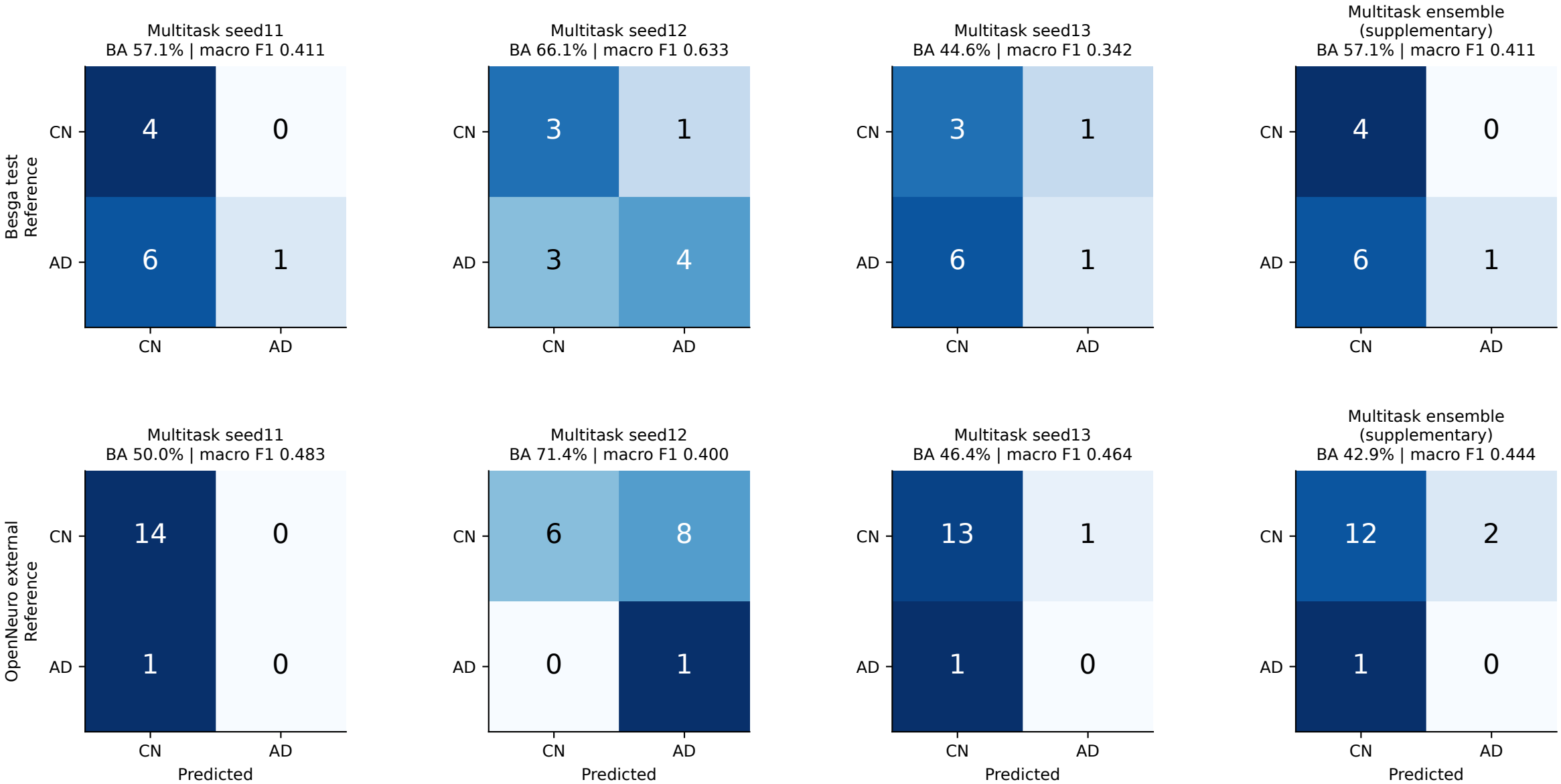
Non versus rest | positive n=27

Very mild versus rest | positive n=27

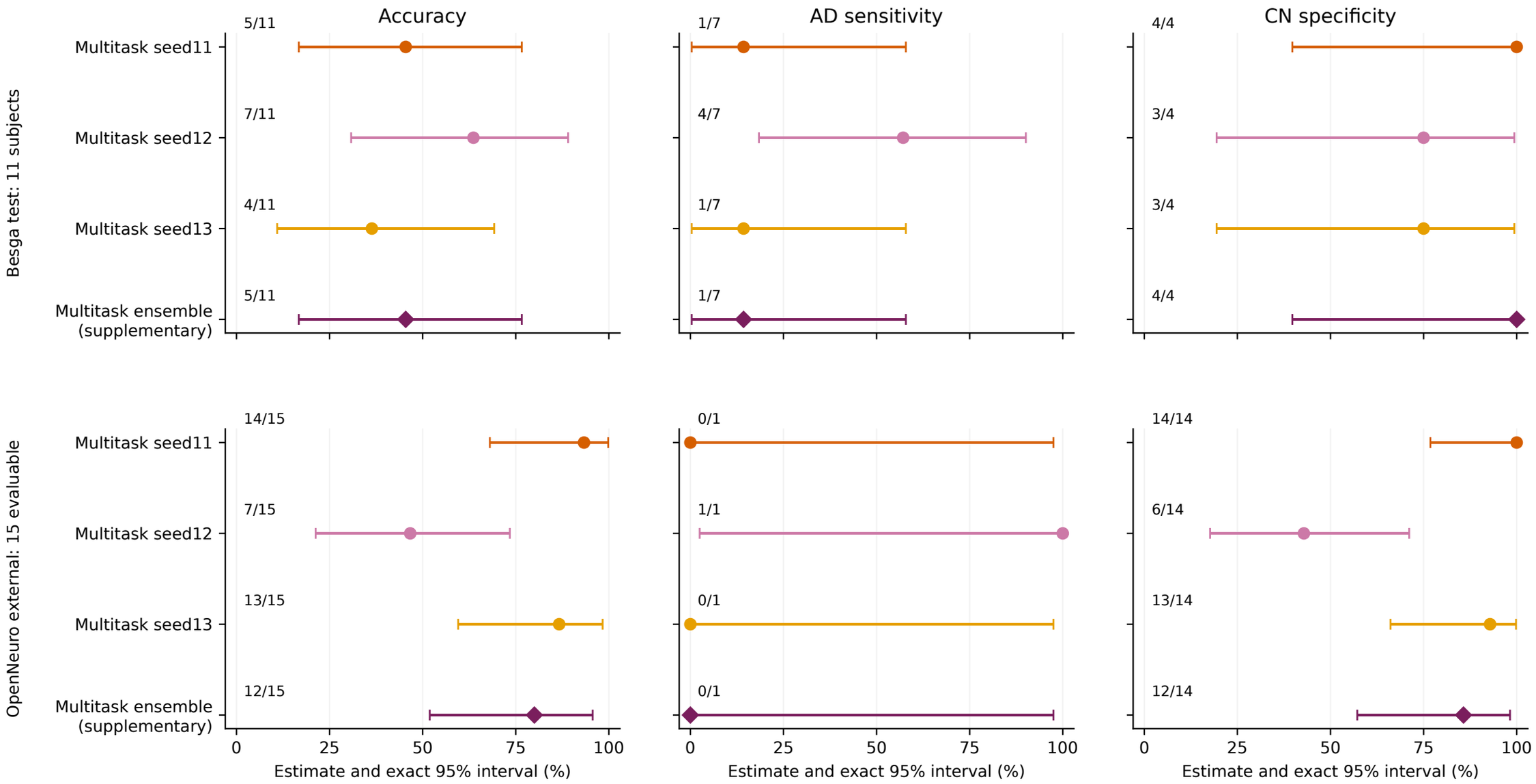
True-positive rate
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0

Old-model exposure: 66 training, 9 validation, 25 test images. No independent clinical claim. Fixed model order; overlapping near-ceiling curves remain in the numeric table. Curves use frozen probabilities, with no threshold selection. Ensembles are supplementary.

Clinical AD/CN | all seeds and fixed ensemble

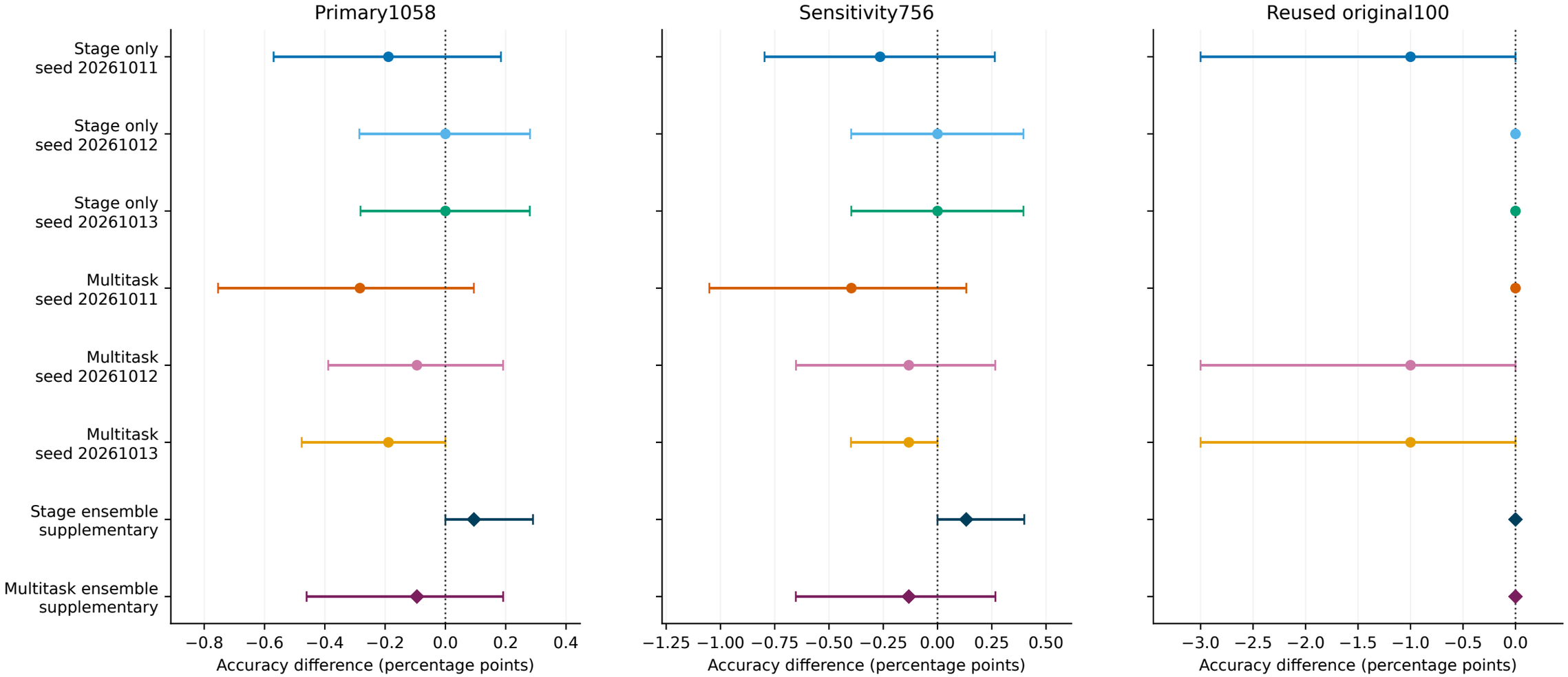


Cells show subject counts; colors show within-reference-class recall. Besga: four CN and seven AD. OpenNeuro: 14 CN and one AD; five MCI excluded. High raw accuracy on this imbalanced pilot can coexist with missing every AD subject. Clinical claims remain exploratory.



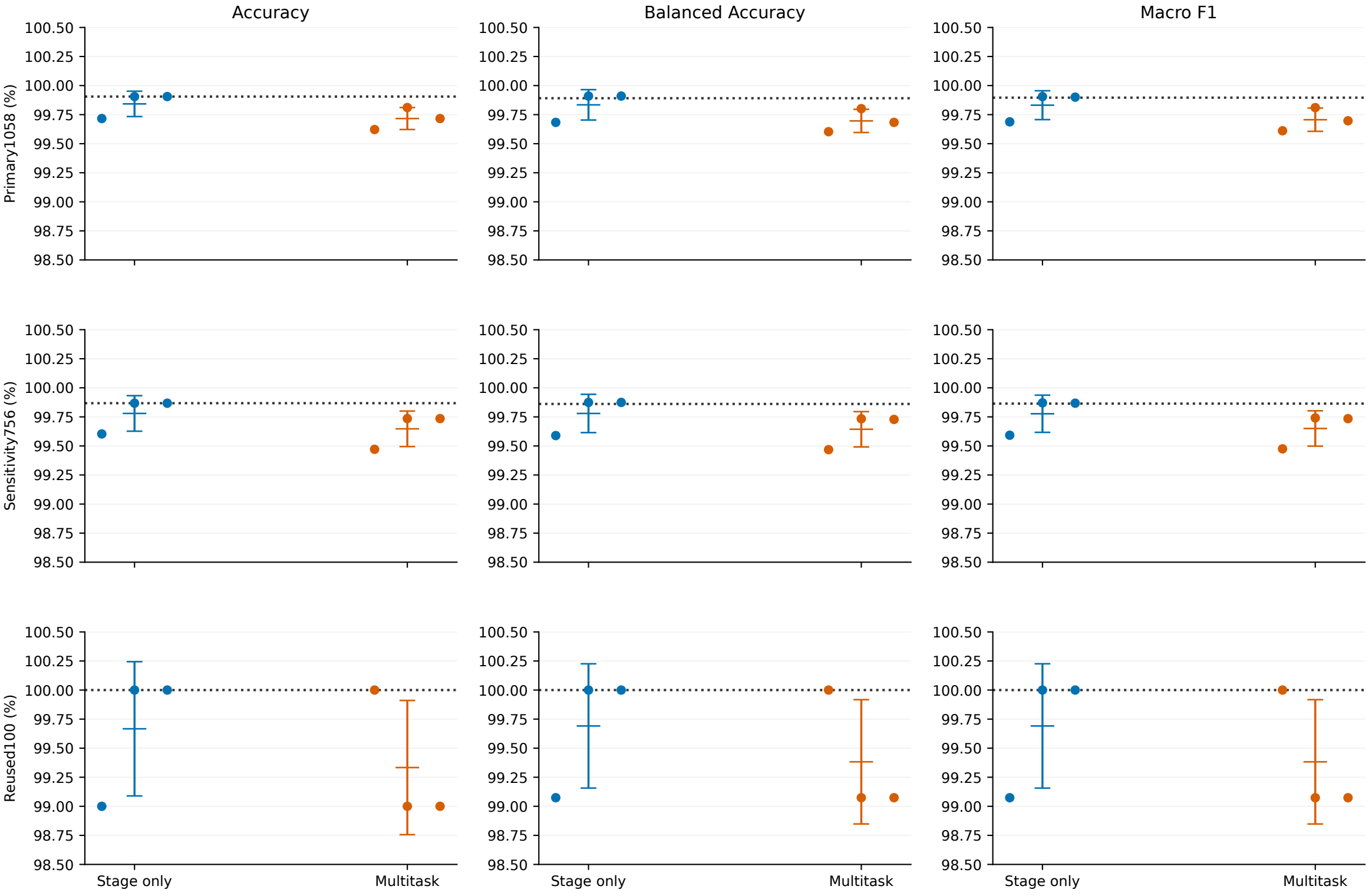
Counts label every estimate. Clopper–Pearson intervals remain very wide: the external cohort has only one AD and 14 CN. Five external MCI subjects are prediction-only. Old and stage-only models have no matching clinical head. The fixed ensemble is supplementary; no model was selected from these outcomes.

Paired accuracy differences from the old baseline



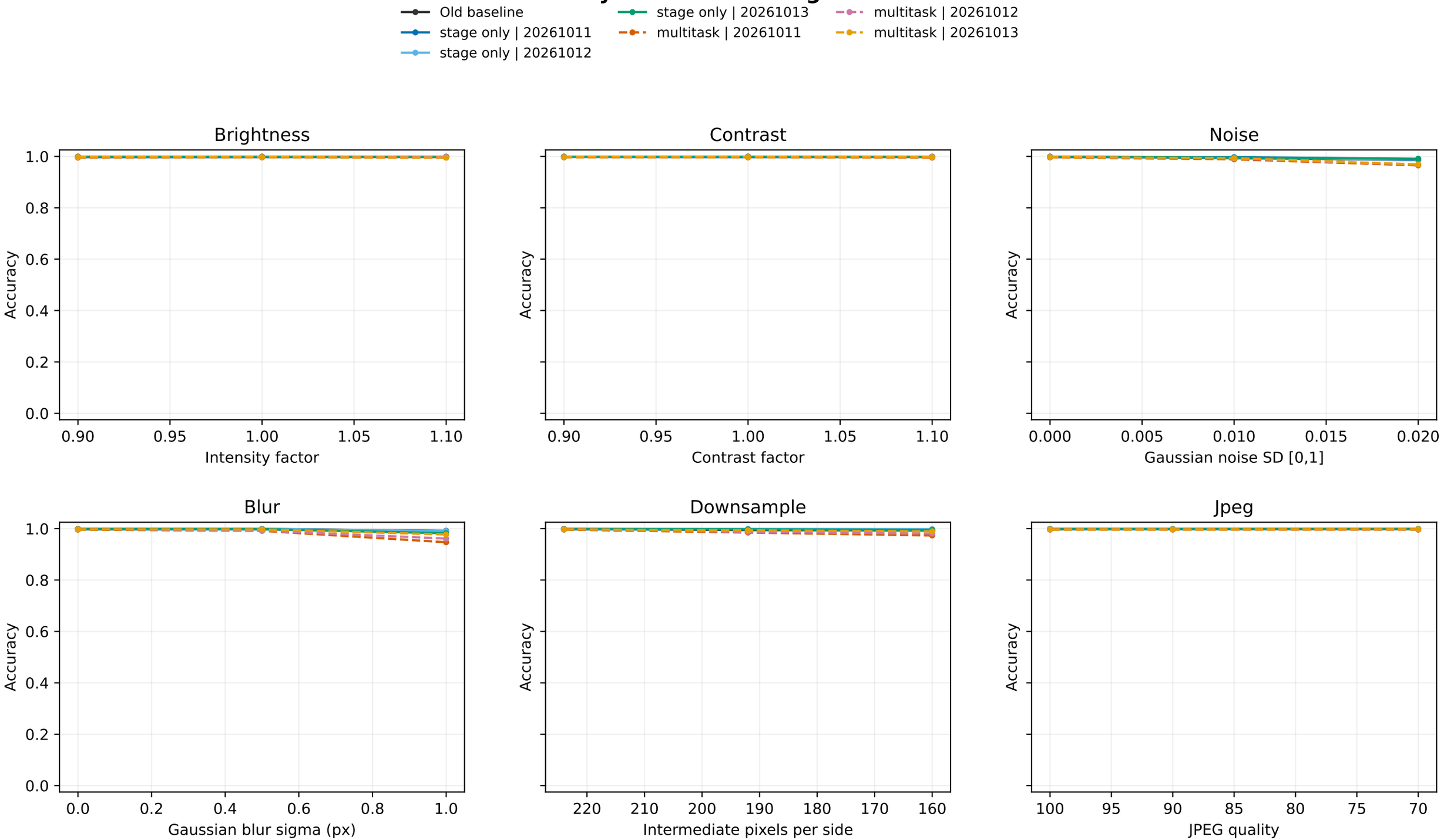
Points and 95% conditional paired bootstrap intervals are copied from frozen analyses (2,000 replicates, no multiplicity adjustment). Primary1058 includes 302 old development group mates; sensitivity756 excludes them. Hash groups are not patients; original100 is reused. Horizontal scales differ; zero-width intervals do not prove equivalence. Diamonds are supplementary ensembles.

All three prespecified seeds | mean and spread



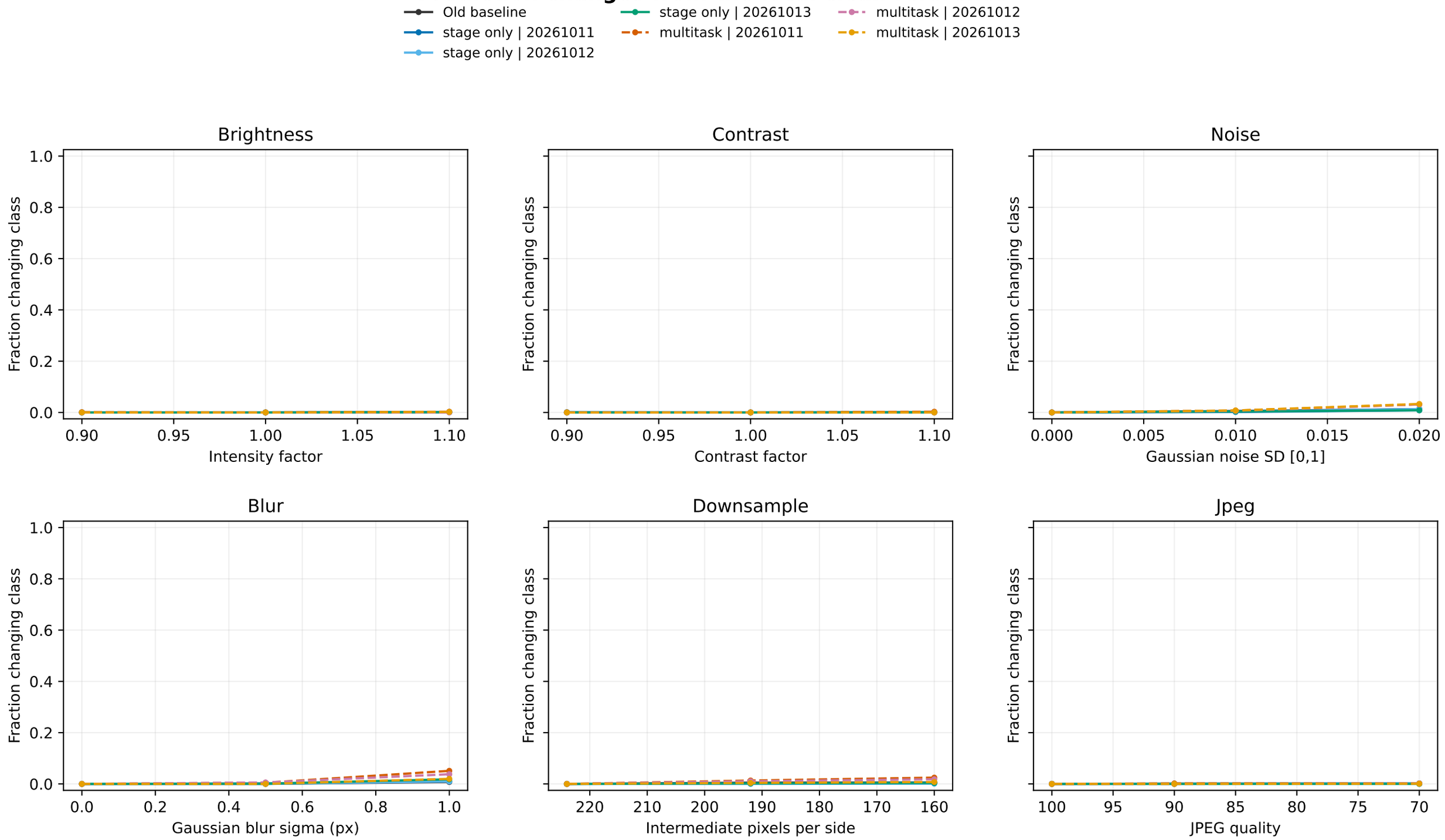
Each dot is one fixed seed; mark is its condition mean and whiskers are sample SD, not confidence intervals (SD bars may exceed 100%). Dotted line: old baseline. Axes show 98.5-100.5%. Primary has old development group exposure; sensitivity groups are not patients; original100 is reused.

POST-HOC DESCRIPTIVE AUDIT | full primary 1058 image rows
Accuracy under fixed image stress



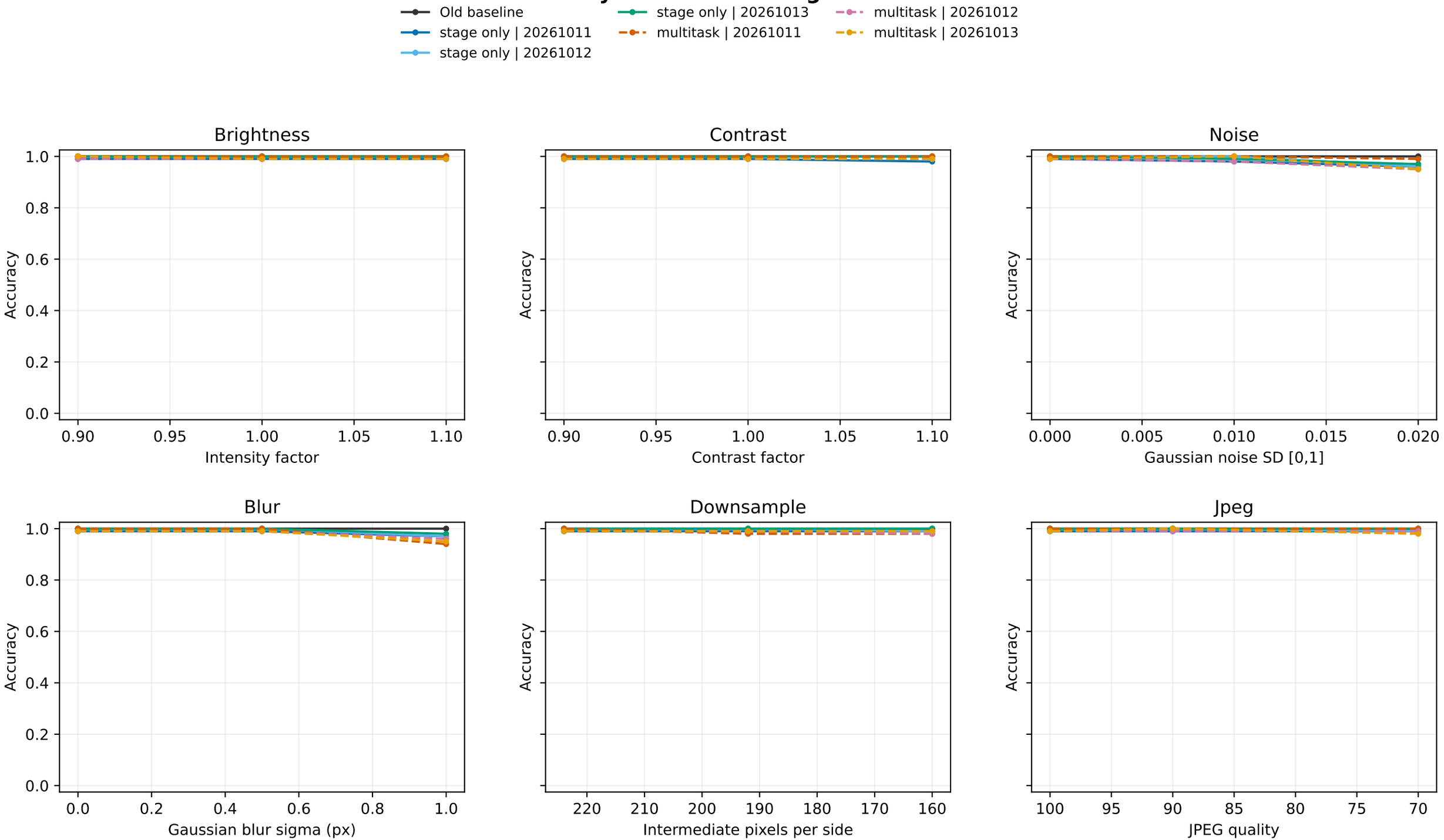
Fixed checkpoints; no tuning or reselection. Synthetic 2D image perturbations do not establish independent patient or clinical robustness.
Lines connect only the measured scenario levels; clean comparisons use the current local device, not a new primary evaluation.

POST-HOC DESCRIPTIVE AUDIT | full primary 1058 image rows
Prediction changes from the local clean reference



Fixed checkpoints; no tuning or reselection. Synthetic 2D image perturbations do not establish independent patient or clinical robustness.
Lines connect only the measured scenario levels; clean comparisons use the current local device, not a new primary evaluation.

POST-HOC DESCRIPTIVE AUDIT | historically reused original100
Accuracy under fixed image stress



Fixed checkpoints; no tuning or reselection. Synthetic 2D image perturbations do not establish independent patient or clinical robustness.
Lines connect only the measured scenario levels; clean comparisons use the current local device, not a new primary evaluation.

POST-HOC DESCRIPTIVE AUDIT | historically reused original100
Prediction changes from the local clean reference

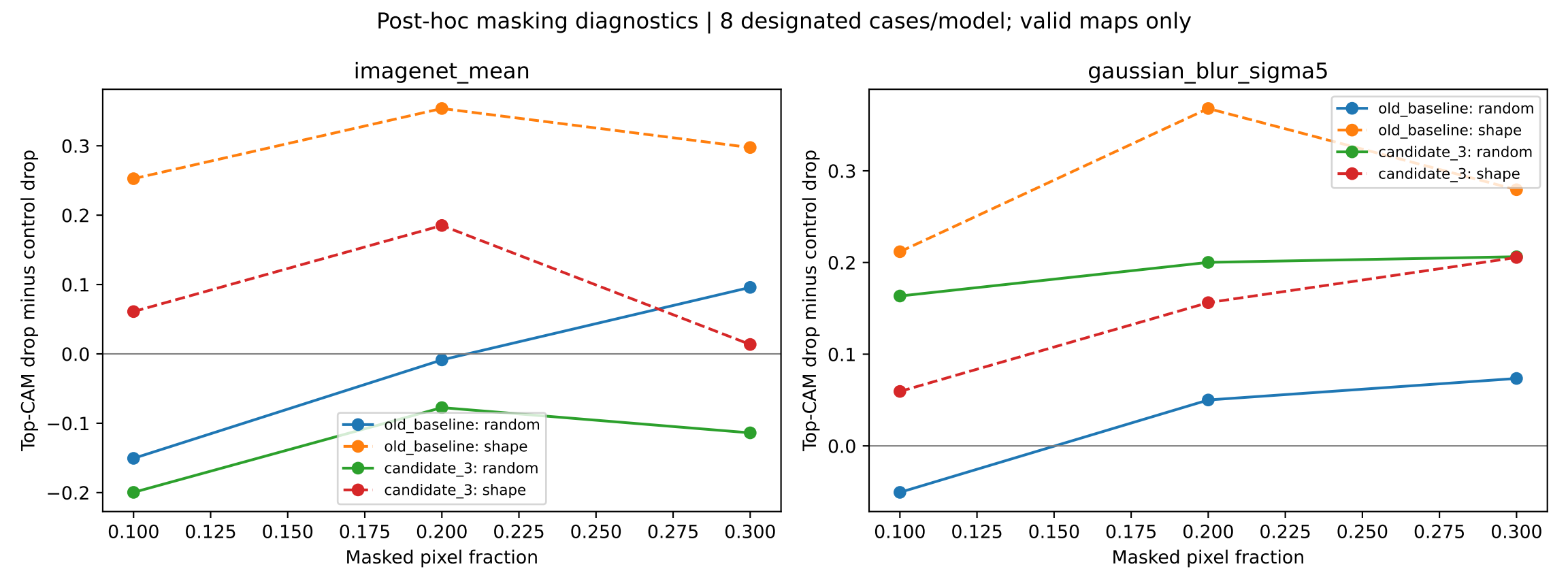
Old baseline
stage only | 20261011
stage only | 20261012
stage only | 20261013
multitask | 20261011
multitask | 20261012
multitask | 20261013

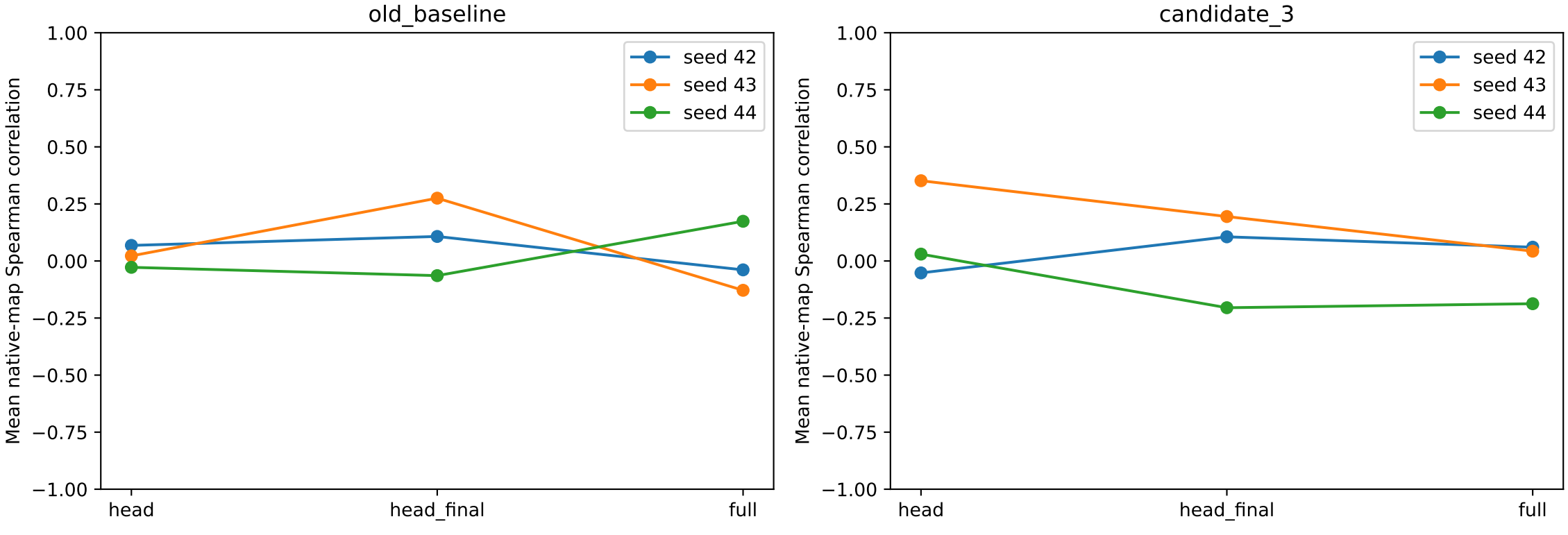
Brightness
Contrast
Noise
Blur
Downsample
Jpeg

Fraction changing class

Intensity factor
Contrast factor
Gaussian noise SD [0,1]
Gaussian blur sigma (px)
Intermediate pixels per side
JPEG quality

Fixed checkpoints; no tuning or reselection. Synthetic 2D image perturbations do not establish independent patient or clinical robustness.
Lines connect only the measured scenario levels; clean comparisons use the current local device, not a new primary evaluation.


Post-hoc masking diagnostics | 8 designated cases/model; valid maps only
imagenet_mean
gaussian_blur_sigma5
Top-CAM drop minus control drop
Masked pixel fraction
old_baseline: random
old_baseline: shape
candidate_3: random
candidate_3: shape
0.3
0.2
0.1
0.0
−0.1
−0.2
0.100
0.125
0.150
0.175
0.200
0.225
0.250
0.275
0.300


Post-hoc parameter randomization | valid maps only; failures recorded
old_baseline
candidate_3
Mean native-map Spearman correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
seed 42
seed 43
seed 44
head
head_final
full